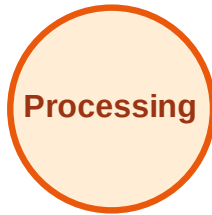


BFS Traversal

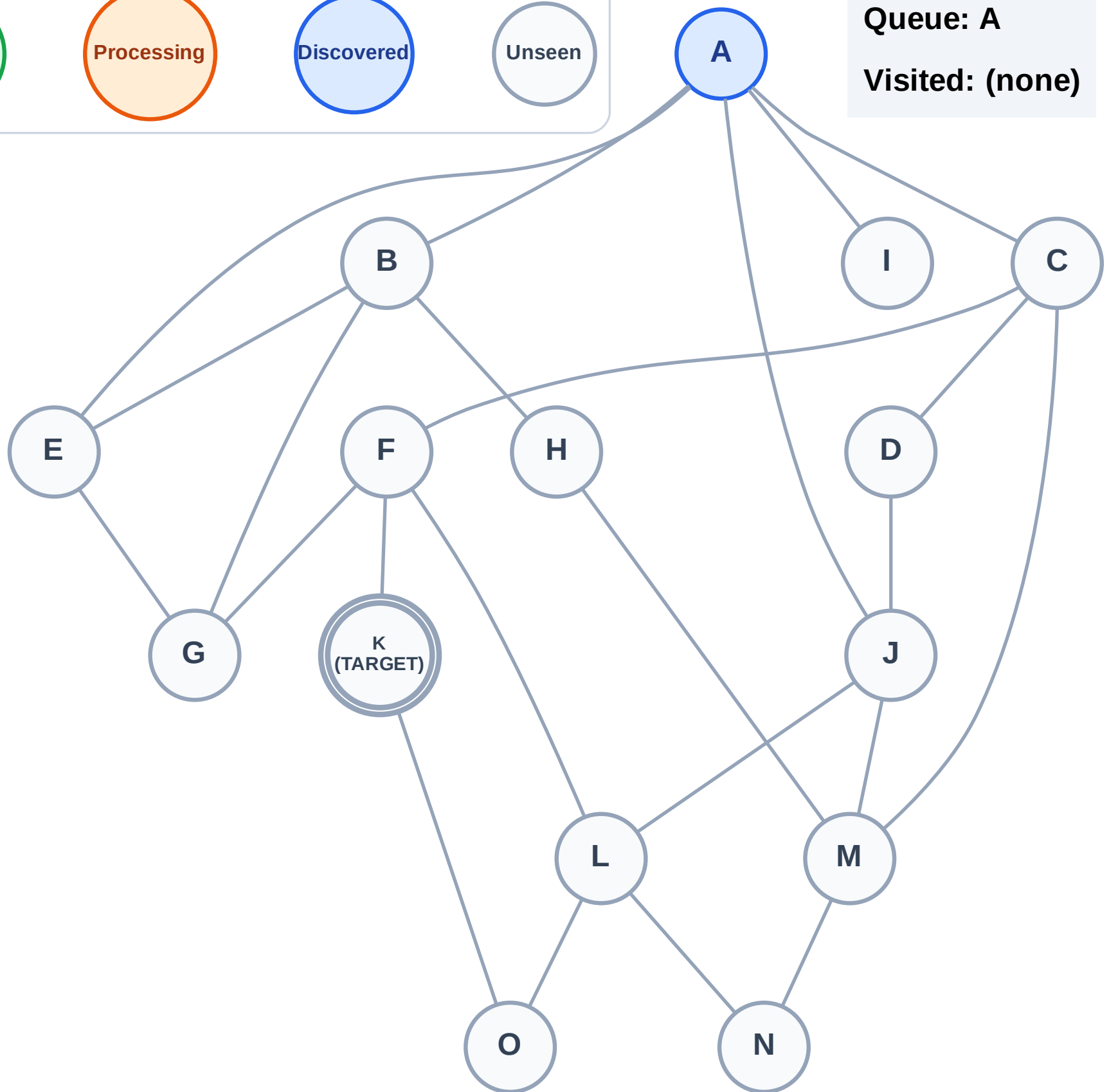
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

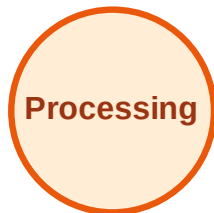
Visited: (none)



BFS Traversal

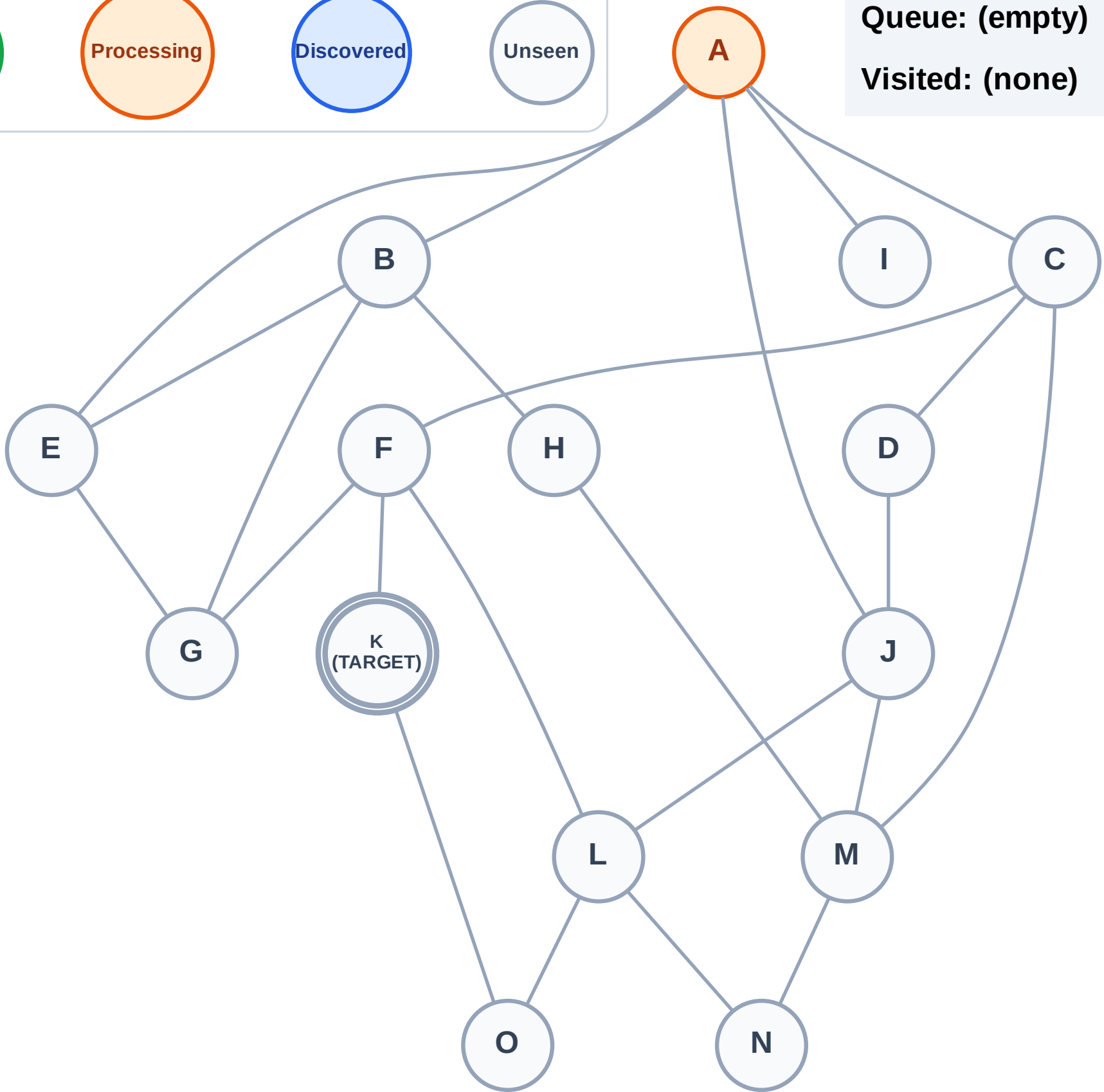
Step 2: Process node A

Legend



Queue: (empty)

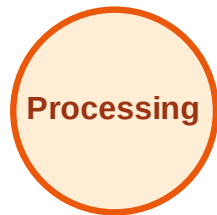
Visited: (none)



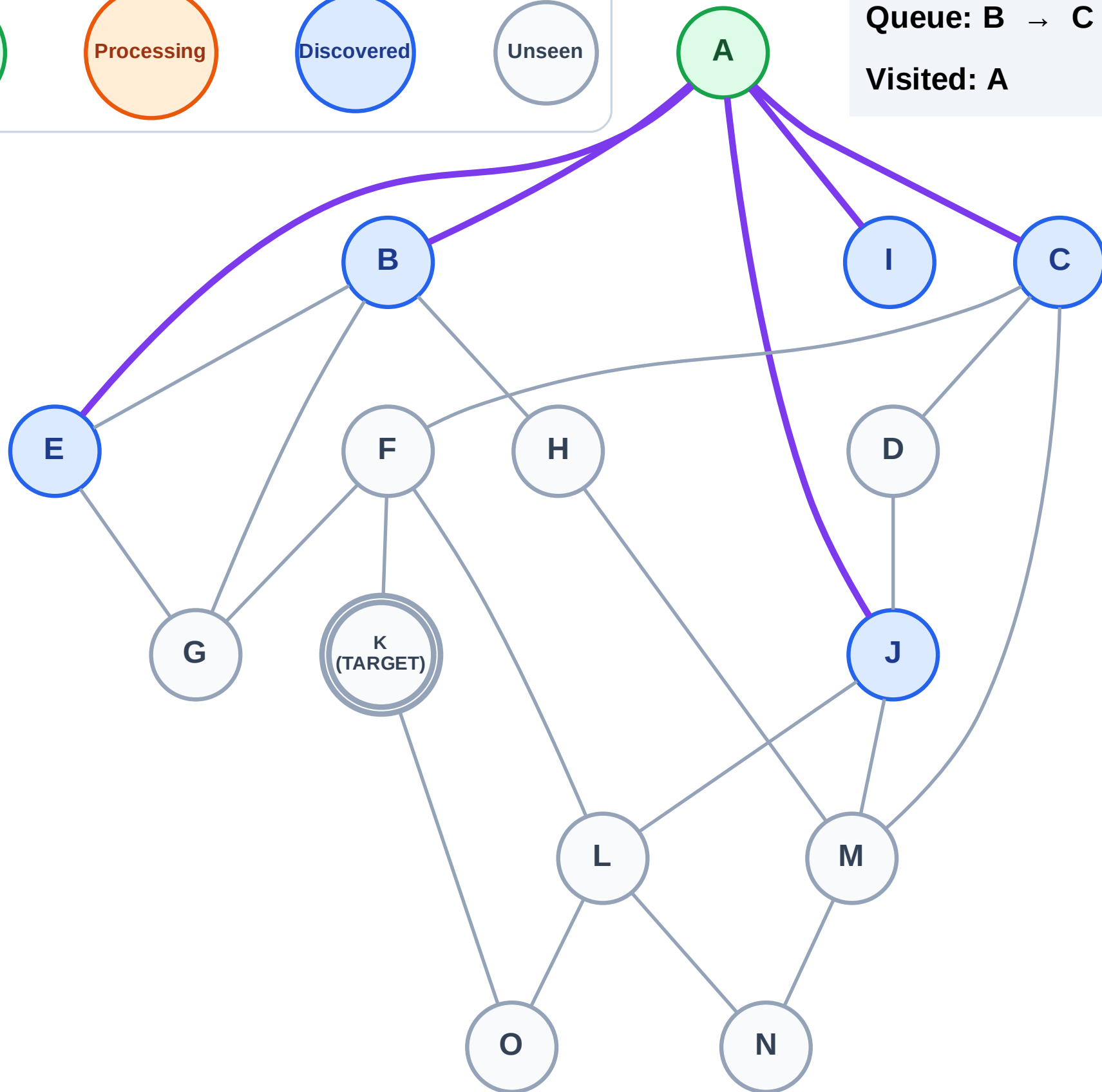
BFS Traversal

Step 3: Discover B, C, E, I, J from A

Legend



Queue: B → C → E → I → J
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

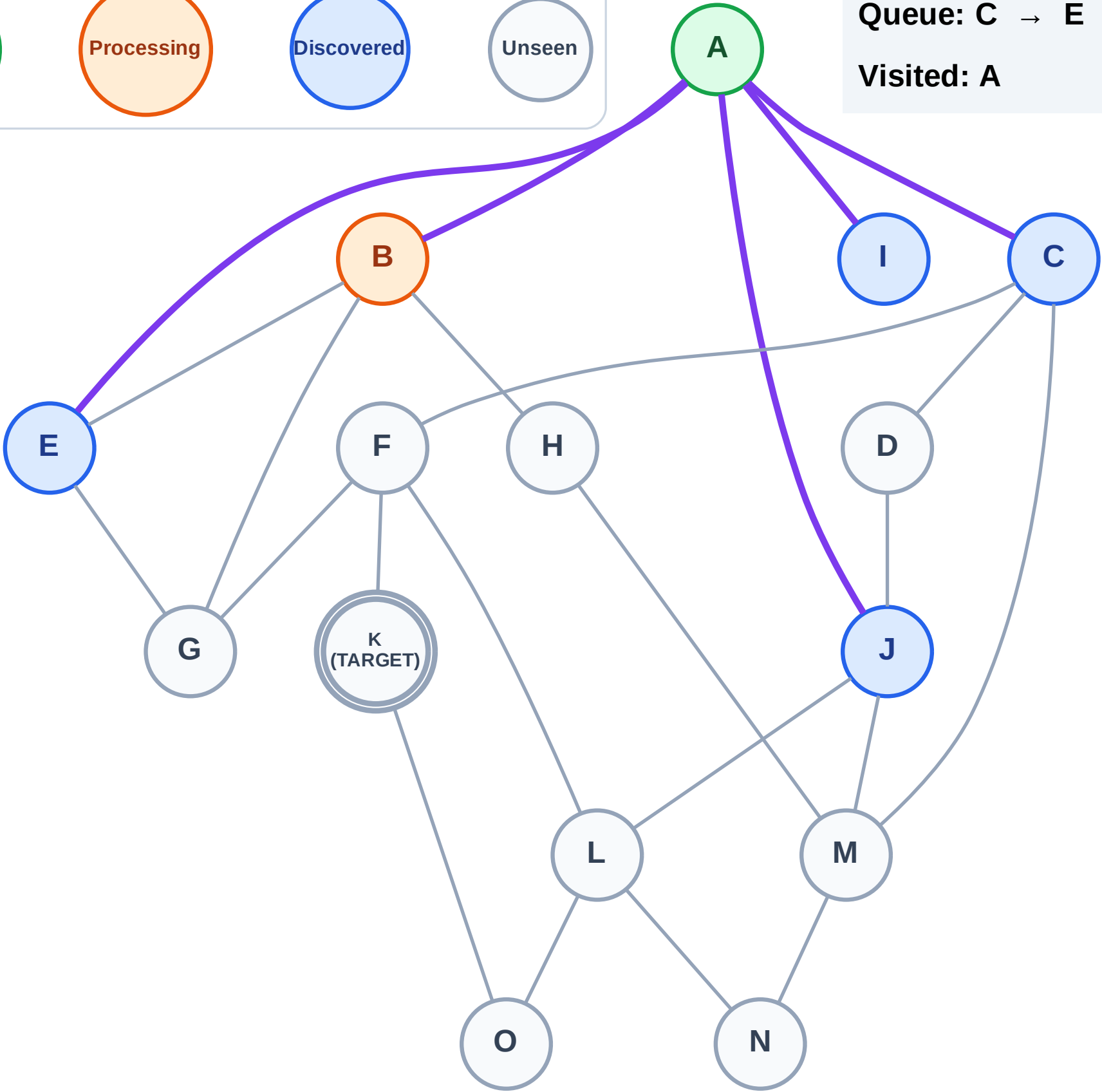
Processing

Discovered

Unseen

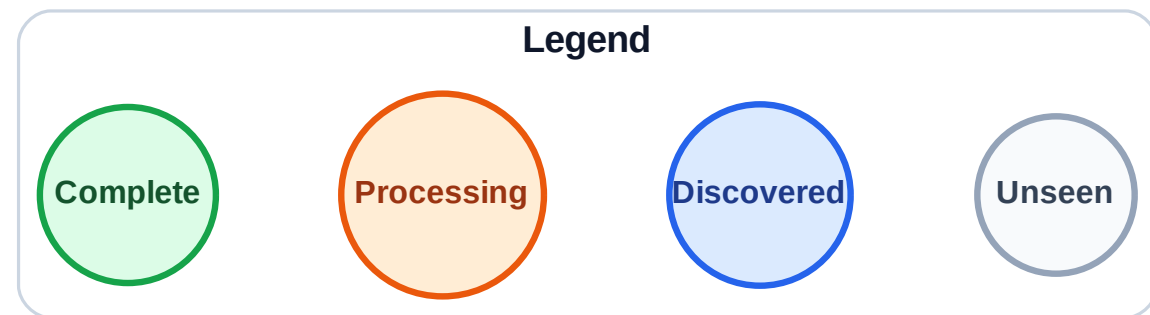
Queue: C → E → I → J

Visited: A

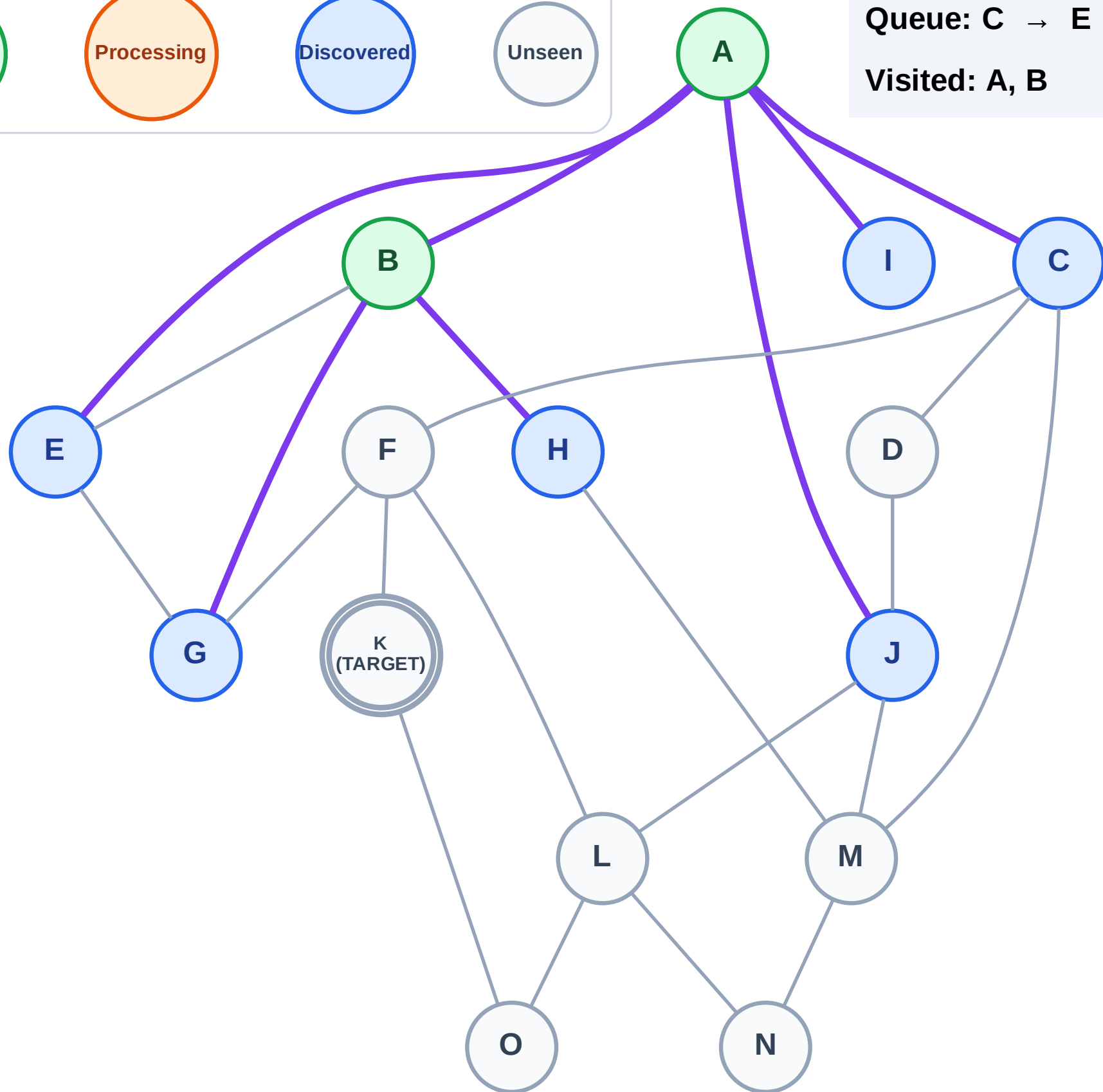


BFS Traversal

Step 5: Discover G, H from B



Queue: C → E → I → J → G → H
Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

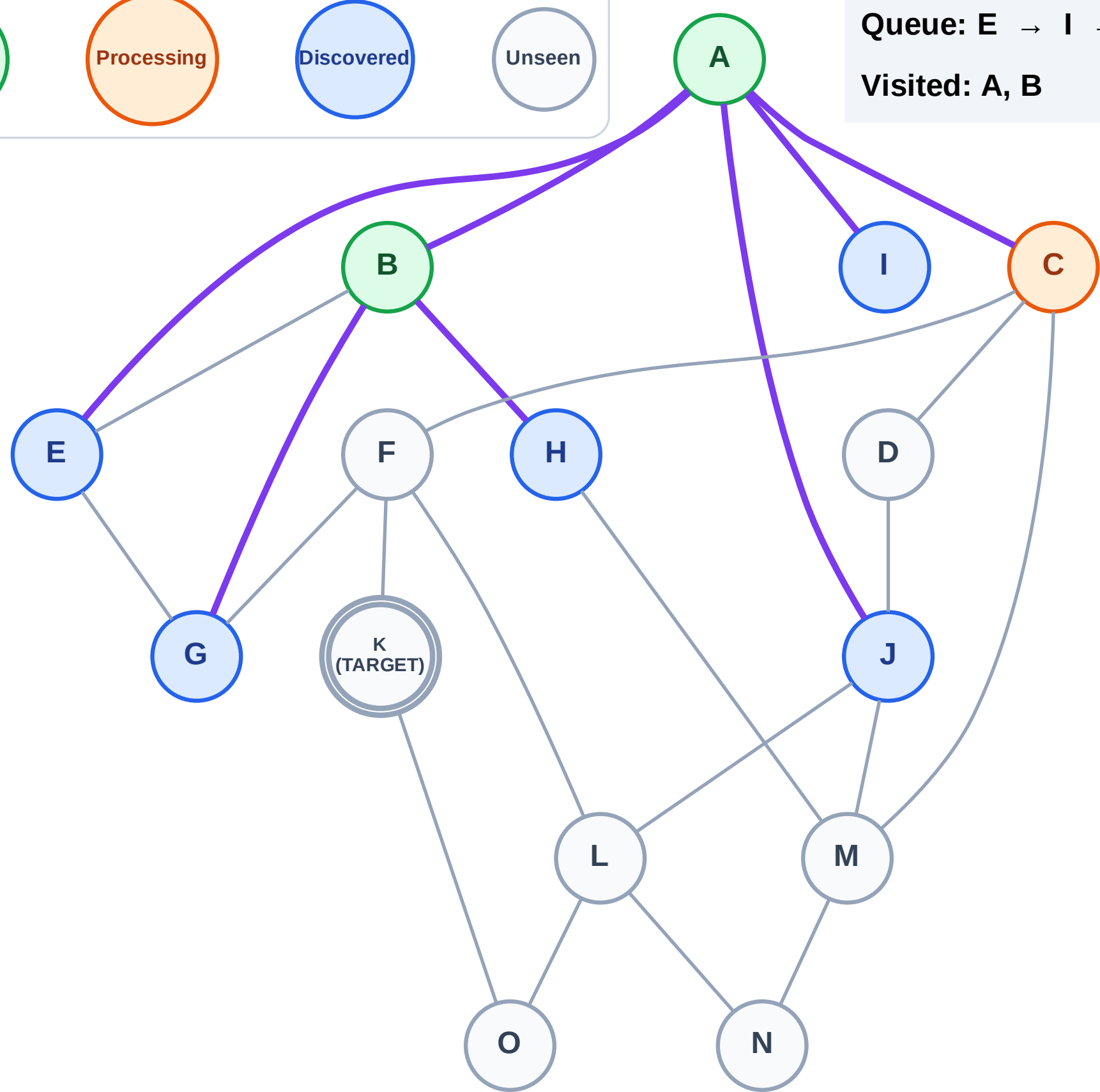
Processing

Discovered

Unseen

Queue: E → I → J → G → H

Visited: A, B



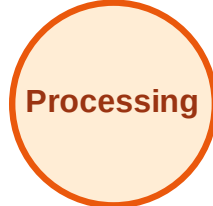
BFS Traversal

Step 7: Discover D, F, M from C

Legend



Complete



Processing



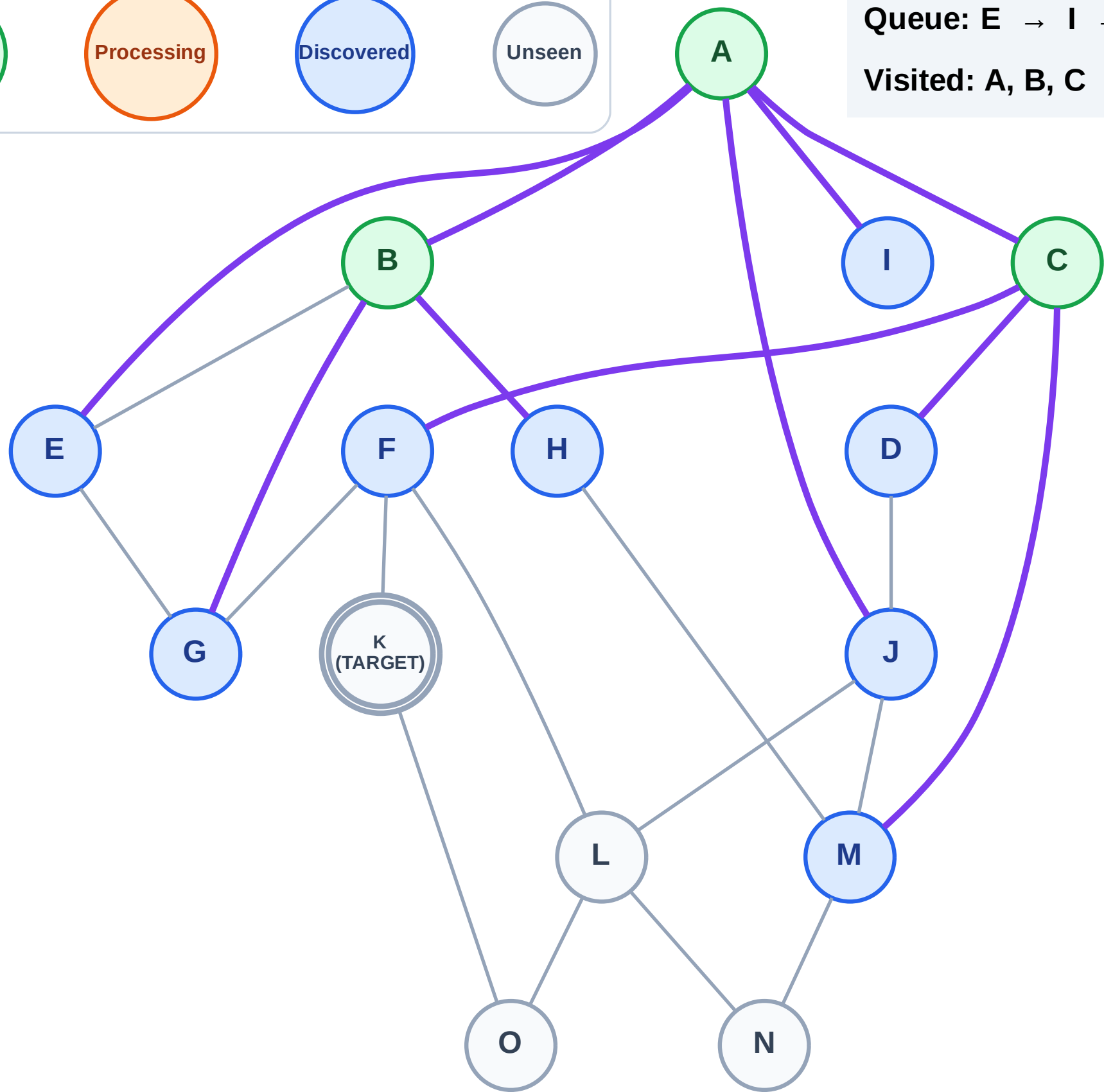
Discovered



Unseen

Queue: E → I → J → G → H → D → F → M

Visited: A, B, C



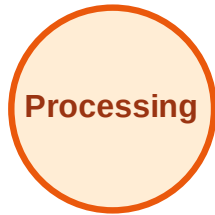
BFS Traversal

Step 8: Process node E

Legend



Complete



Processing



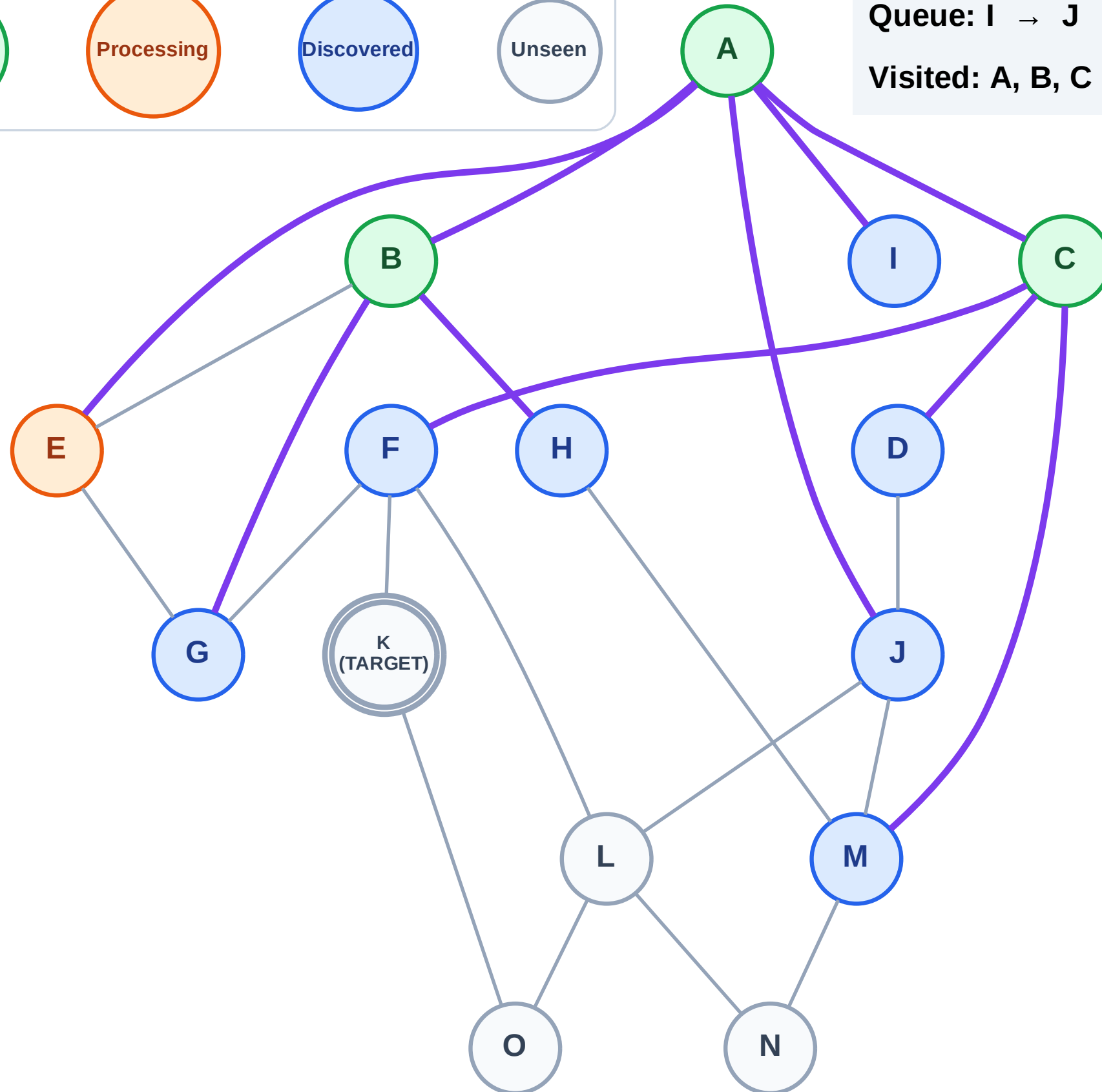
Discovered



Unseen

Queue: I → J → G → H → D → F → M

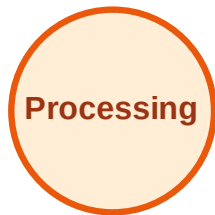
Visited: A, B, C



BFS Traversal

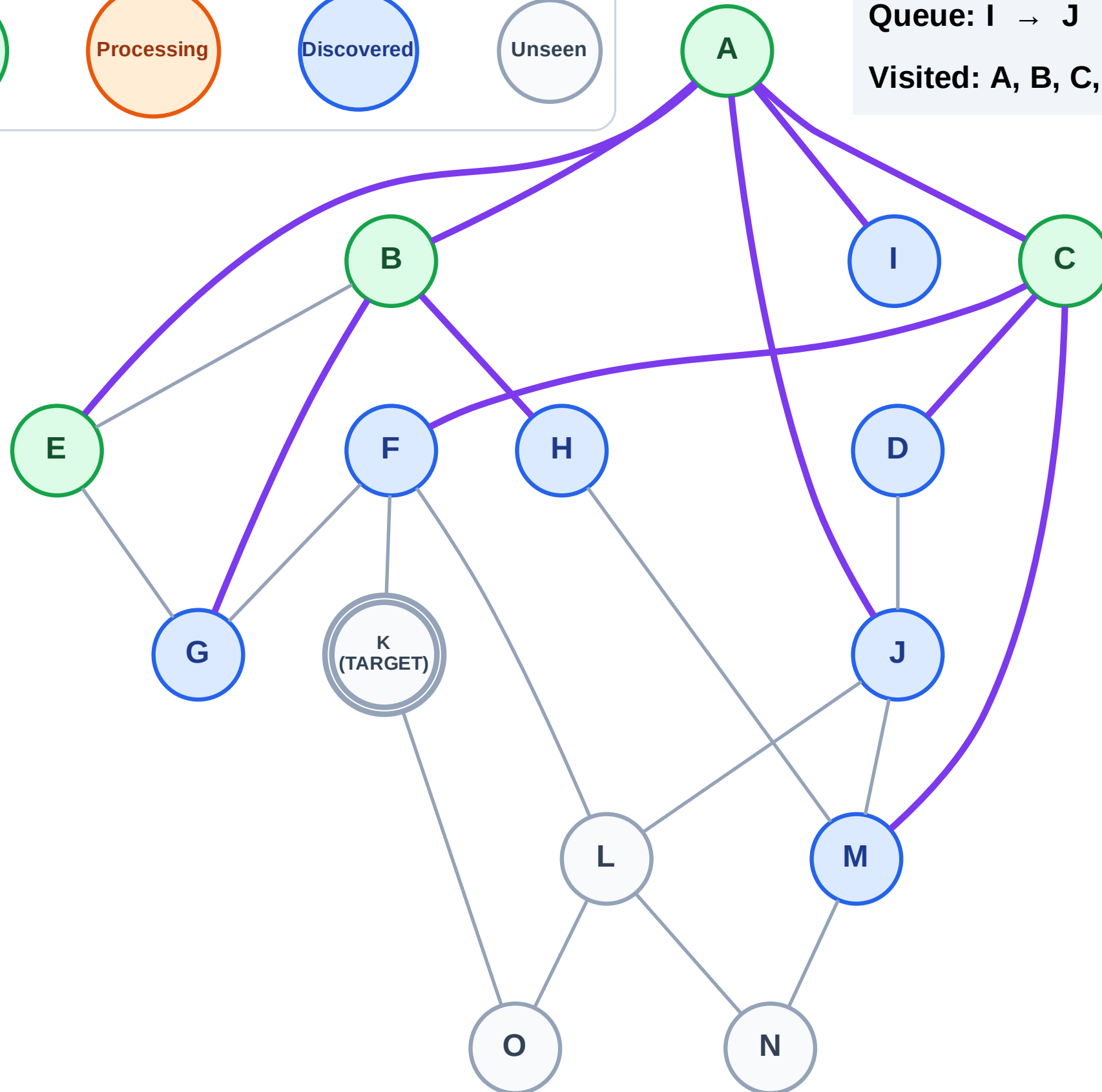
Step 9: No new neighbors from E

Legend



Queue: I → J → G → H → D → F → M

Visited: A, B, C, E



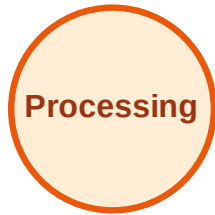
BFS Traversal

Step 10: Process node I

Legend



Complete



Processing



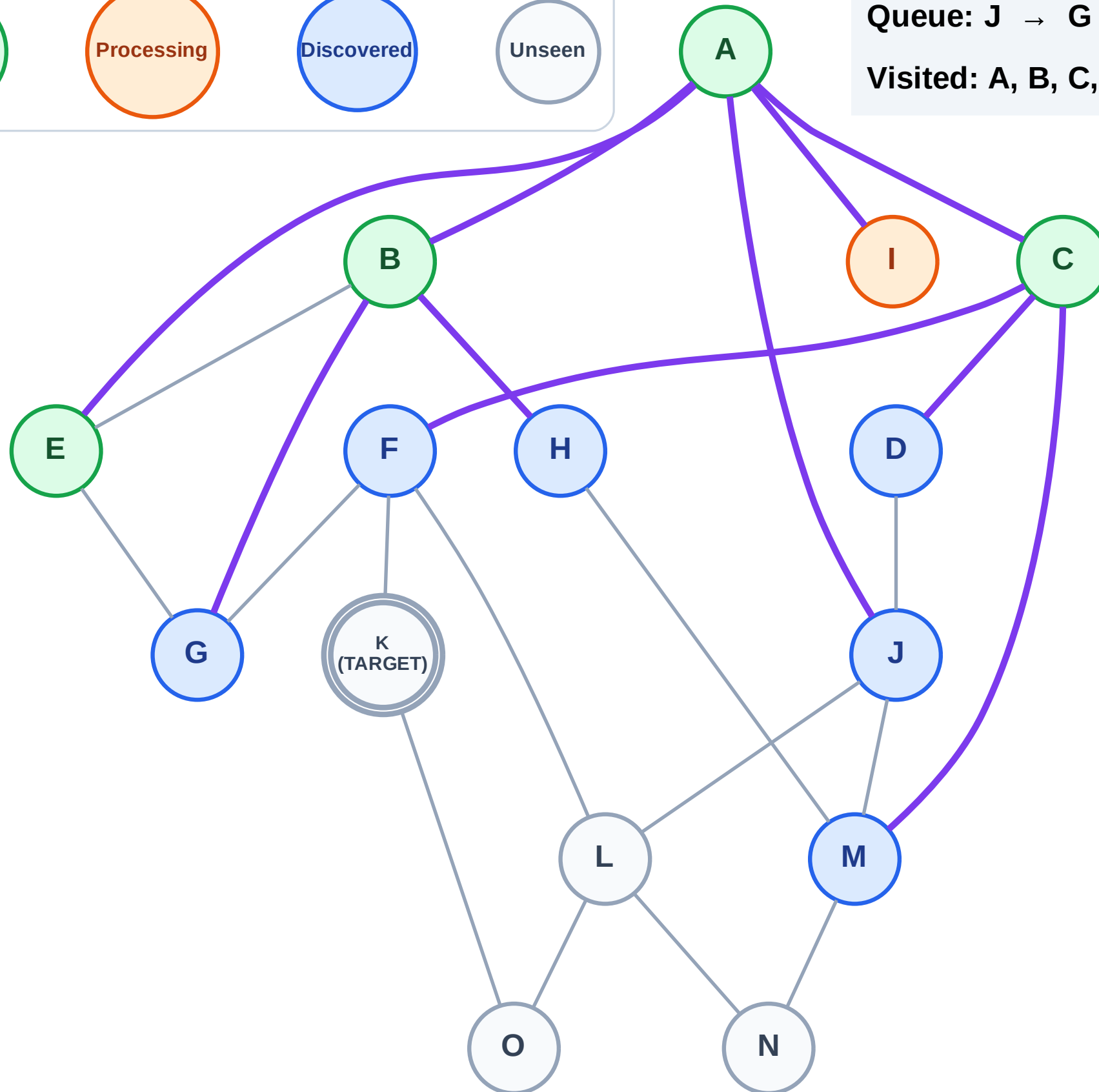
Discovered



Unseen

Queue: J → G → H → D → F → M

Visited: A, B, C, E



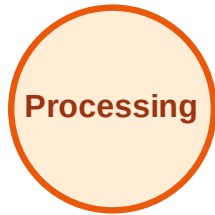
BFS Traversal

Step 11: No new neighbors from I

Legend



Complete



Processing



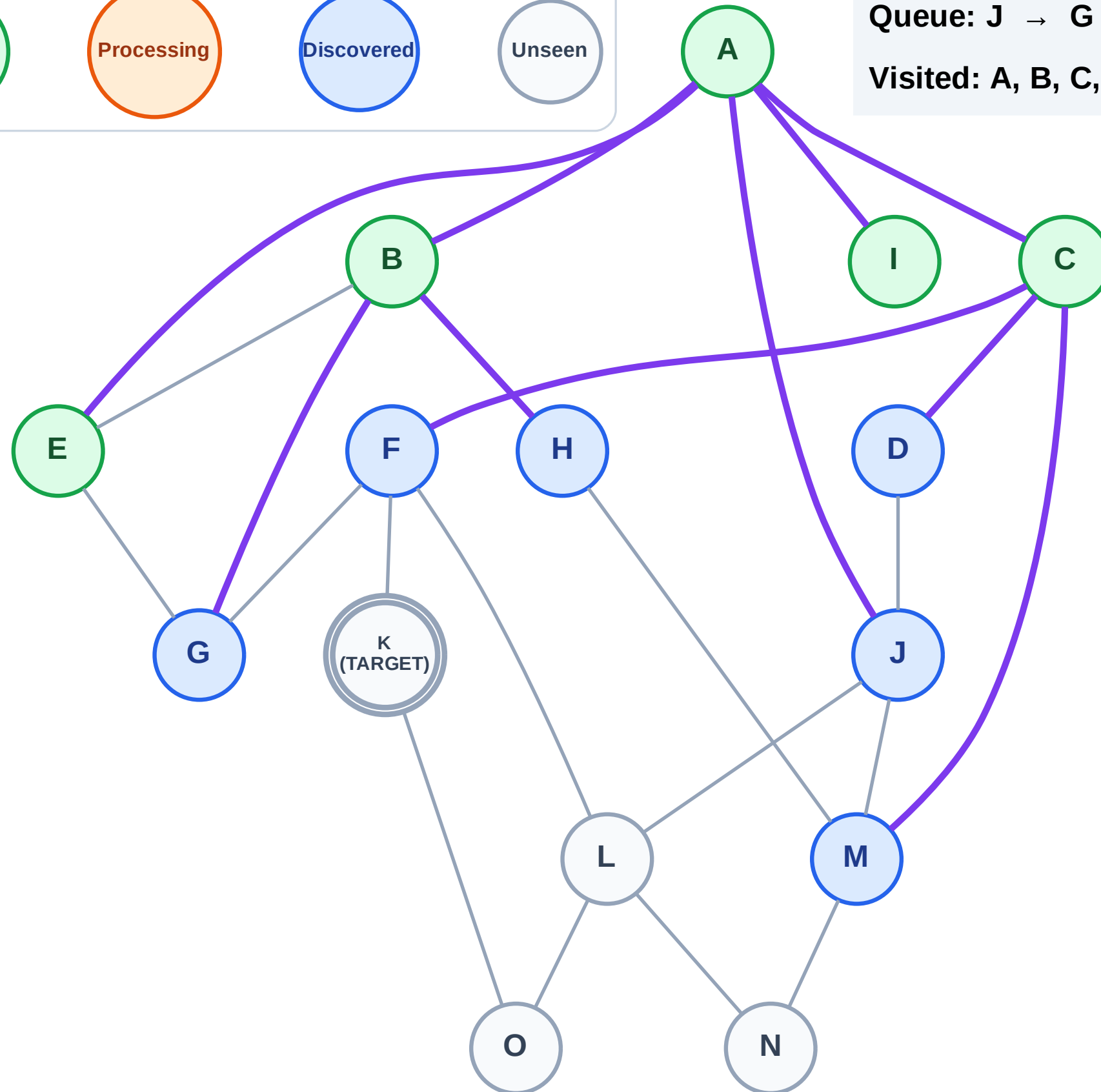
Discovered



Unseen

Queue: J → G → H → D → F → M

Visited: A, B, C, E, I



BFS Traversal

Step 12: Process node J

Legend

Complete

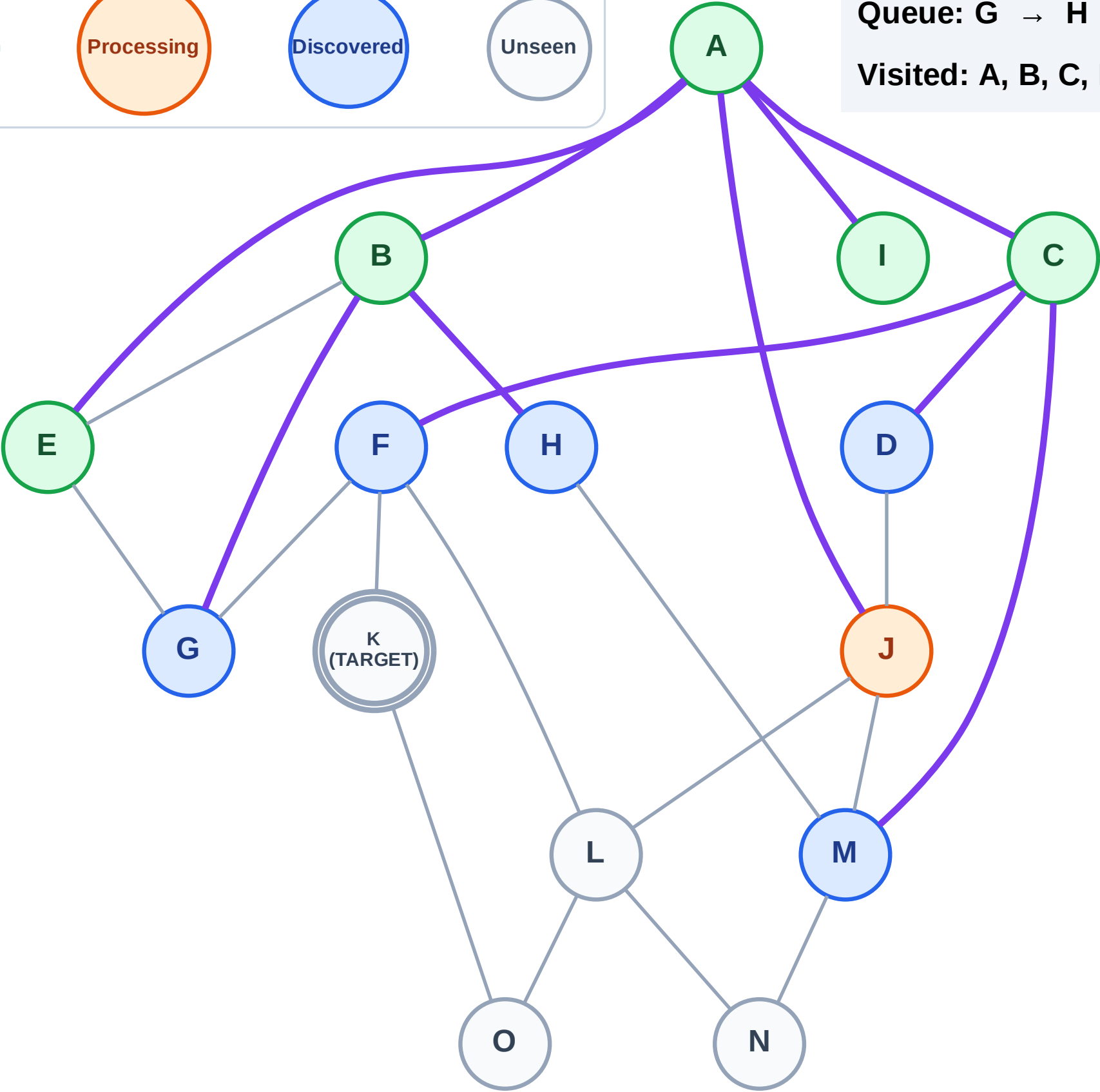
Processing

Discovered

Unseen

Queue: G → H → D → F → M

Visited: A, B, C, E, I



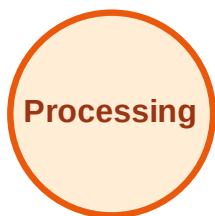
BFS Traversal

Step 13: Discover L from J

Legend



Complete



Processing



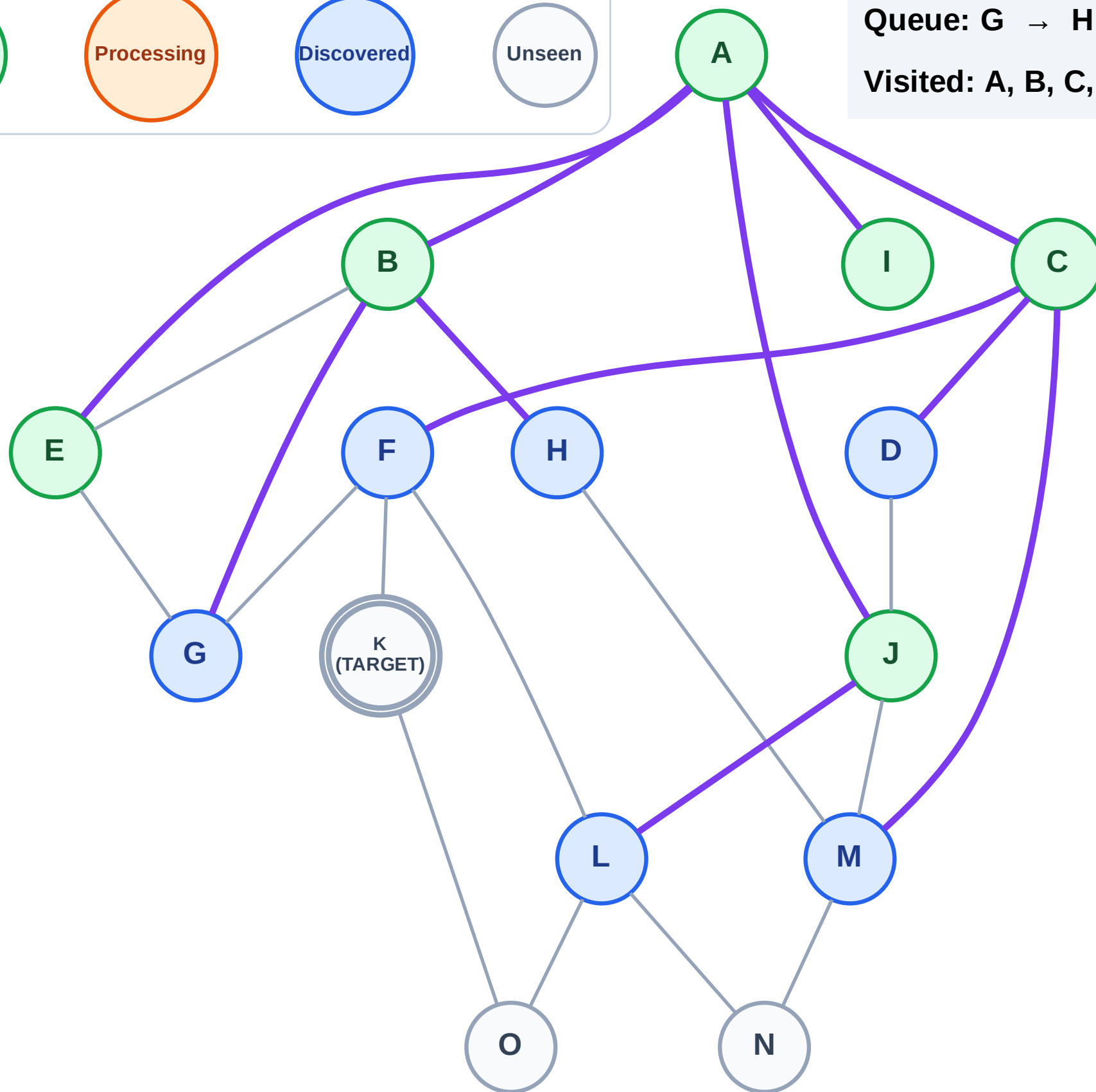
Discovered



Unseen

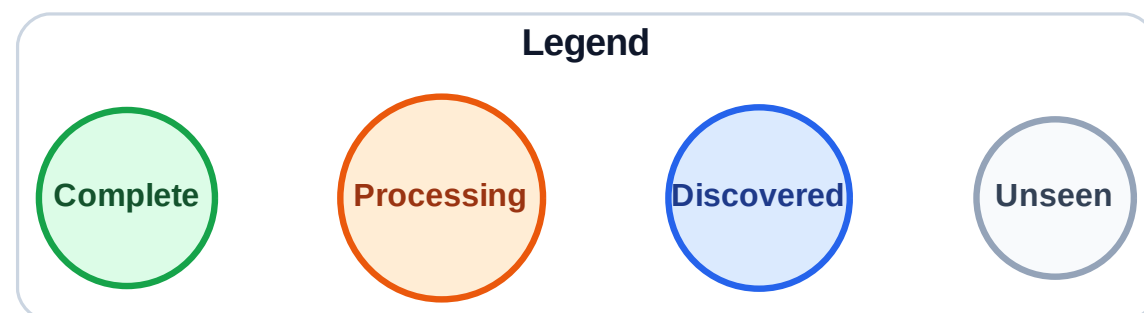
Queue: G → H → D → F → M → L

Visited: A, B, C, E, I, J

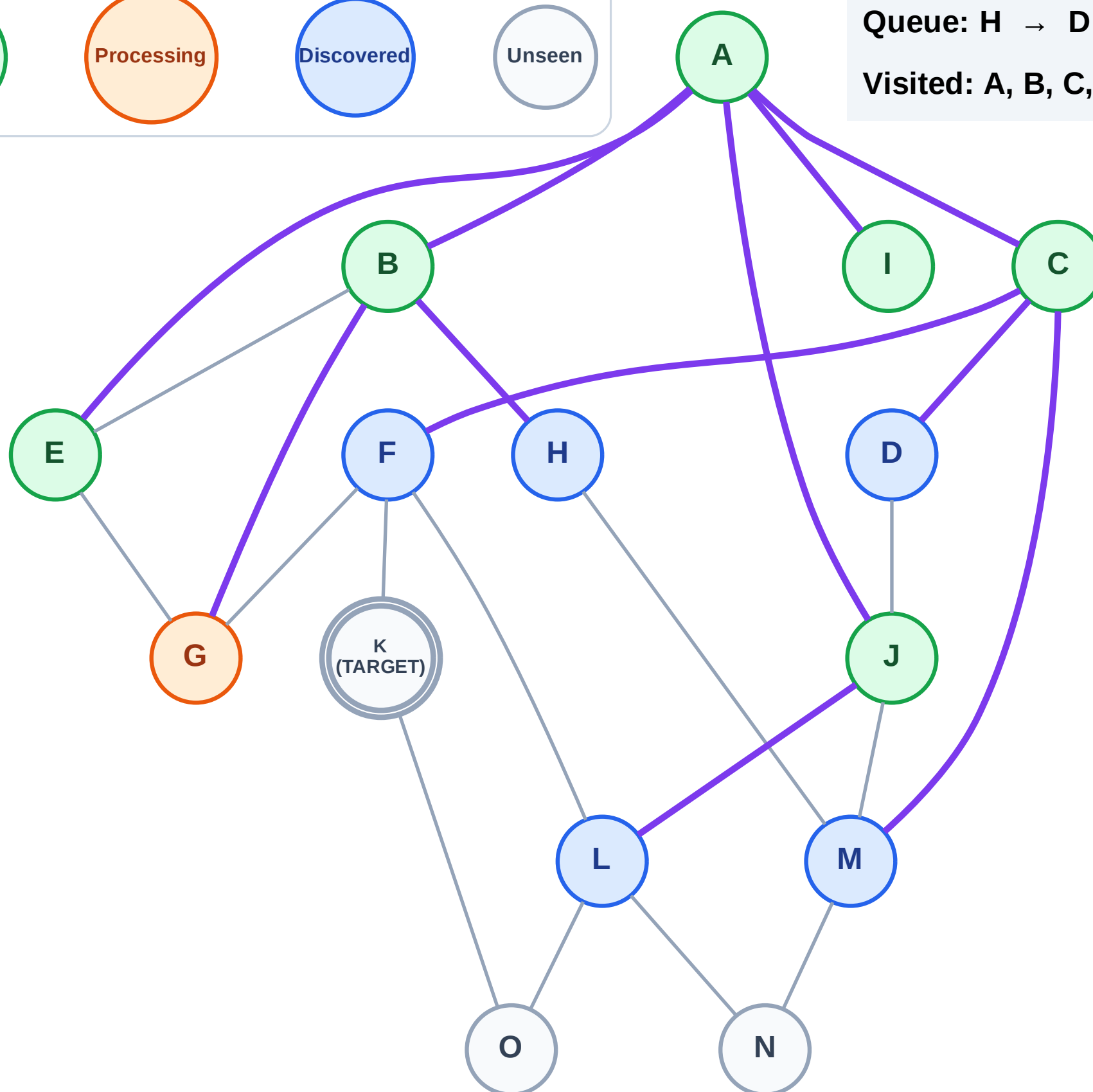


BFS Traversal

Step 14: Process node G



Queue: H → D → F → M → L
Visited: A, B, C, E, I, J



BFS Traversal

Step 15: No new neighbors from G

Legend

Complete

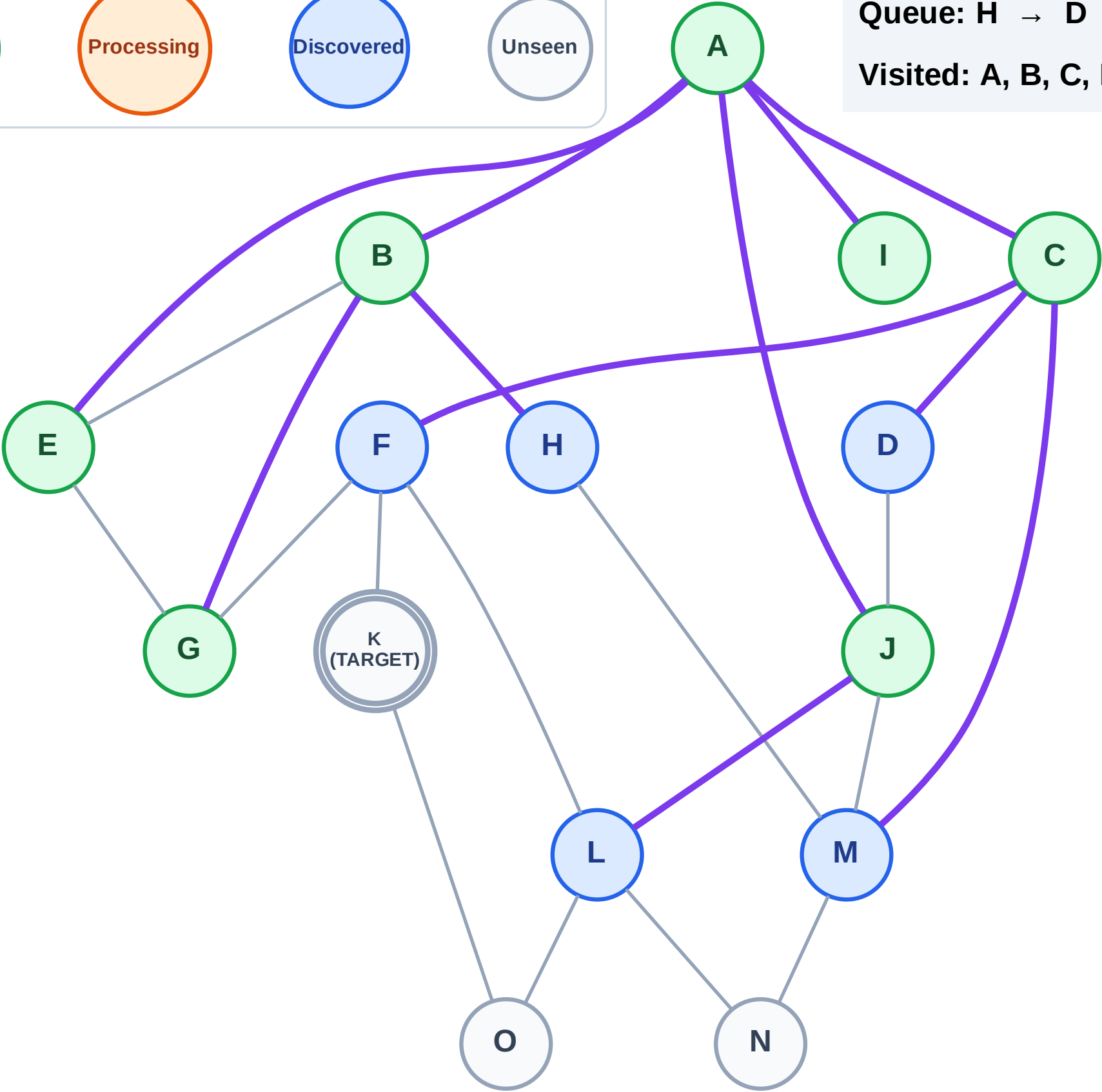
Processing

Discovered

Unseen

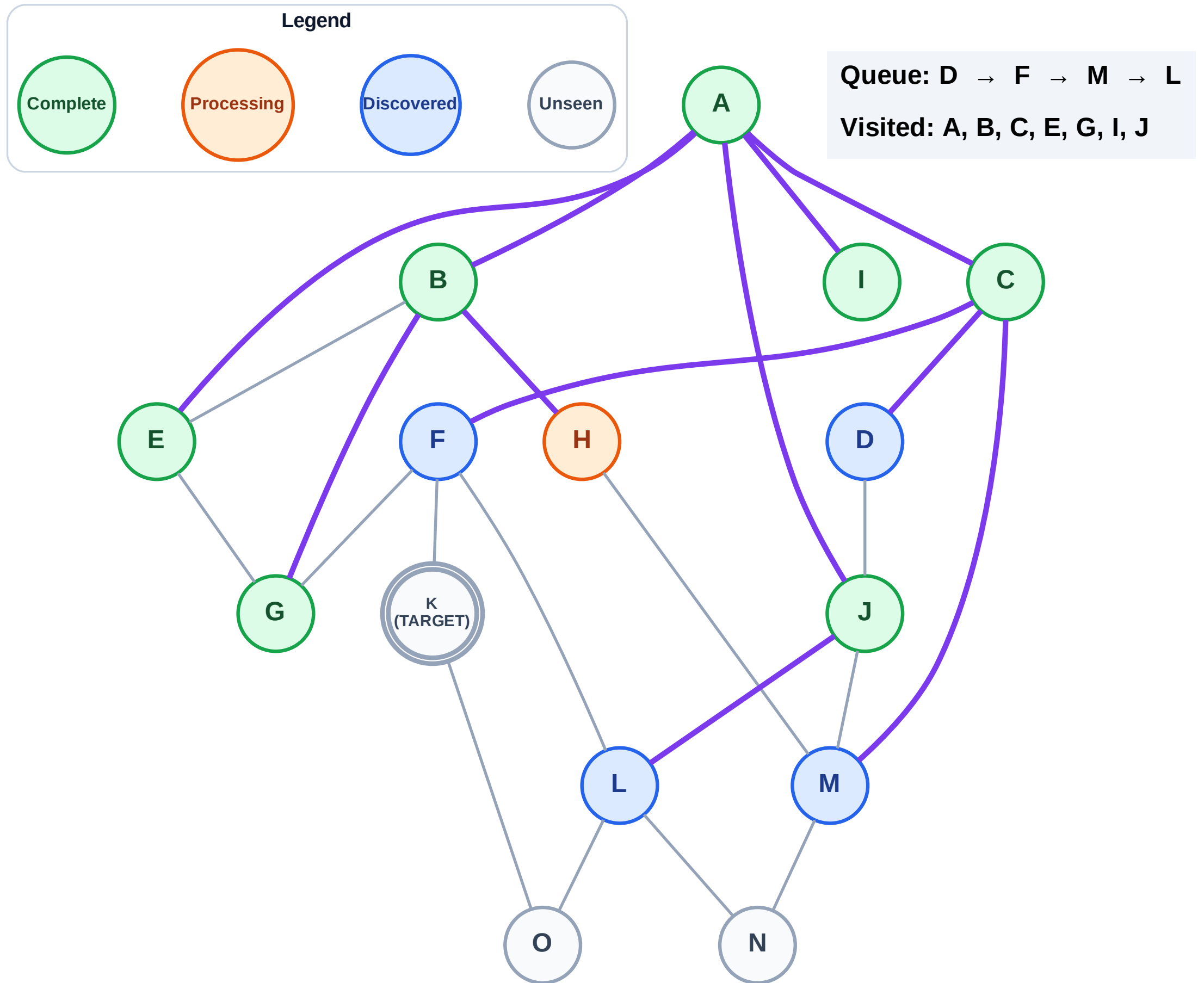
Queue: H → D → F → M → L

Visited: A, B, C, E, G, I, J



Step 16: Process node H

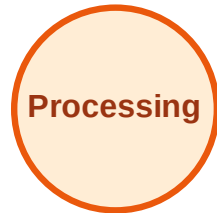
Step 16: Process node H



BFS Traversal

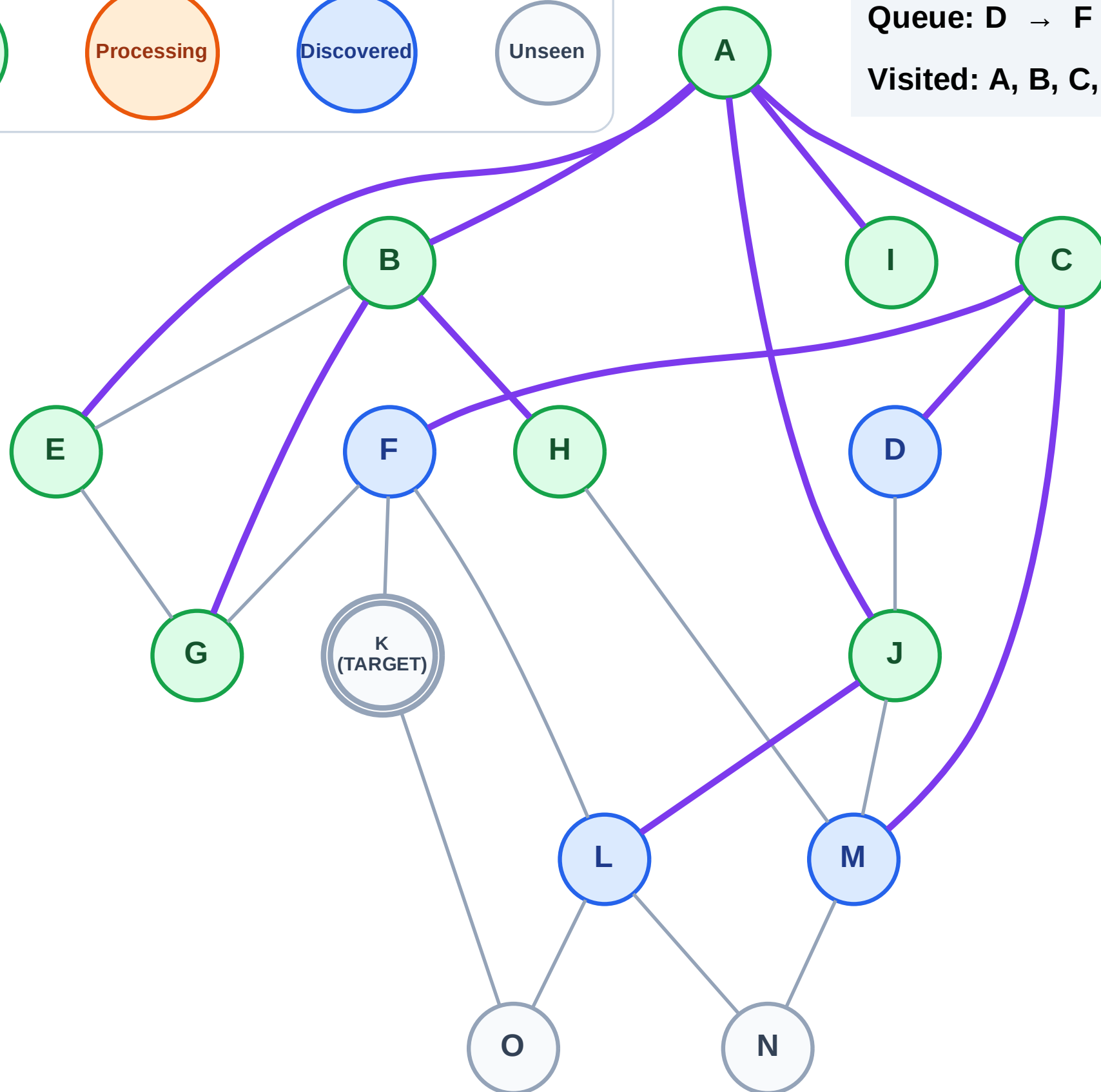
Step 17: No new neighbors from H

Legend



Queue: D → F → M → L

Visited: A, B, C, E, G, H, I, J



BFS Traversal

Step 18: Process node D

Legend

Complete

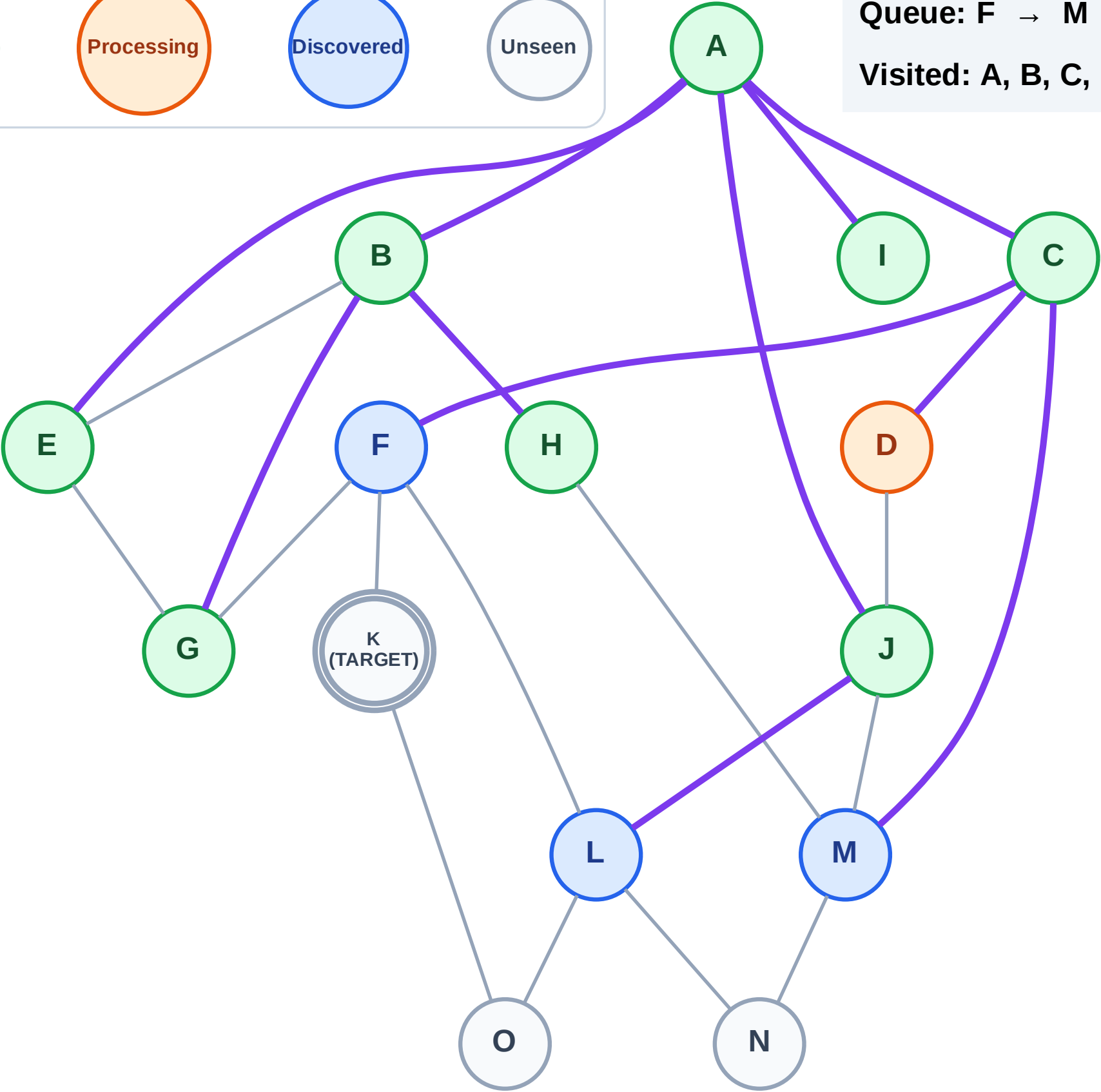
Processing

Discovered

Unseen

Queue: F → M → L

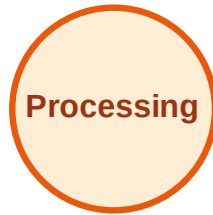
Visited: A, B, C, E, G, H, I, J



BFS Traversal

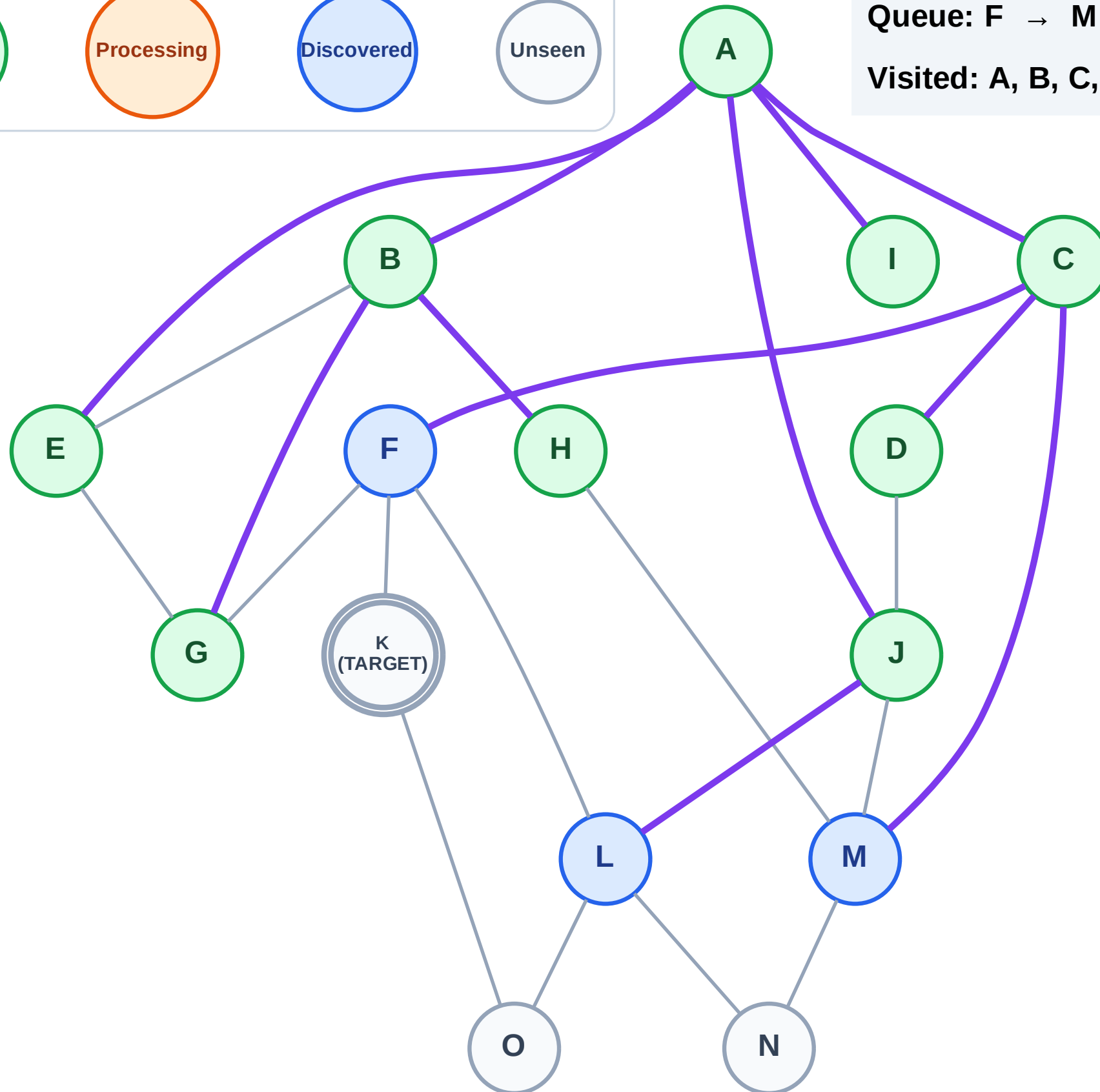
Step 19: No new neighbors from D

Legend



Queue: F → M → L

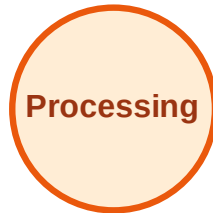
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

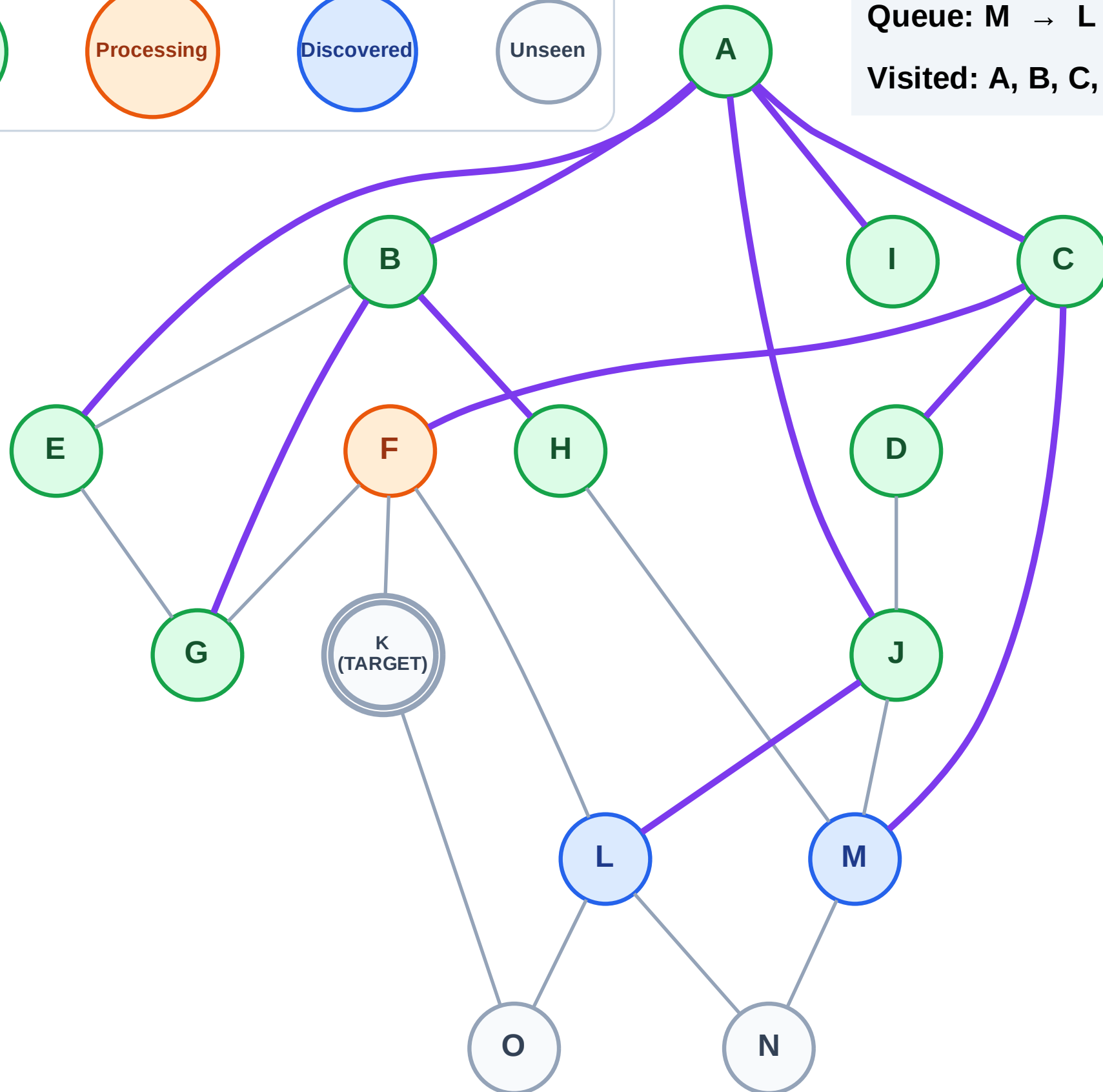
Step 20: Process node F

Legend



Queue: M → L

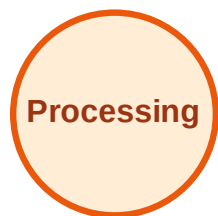
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

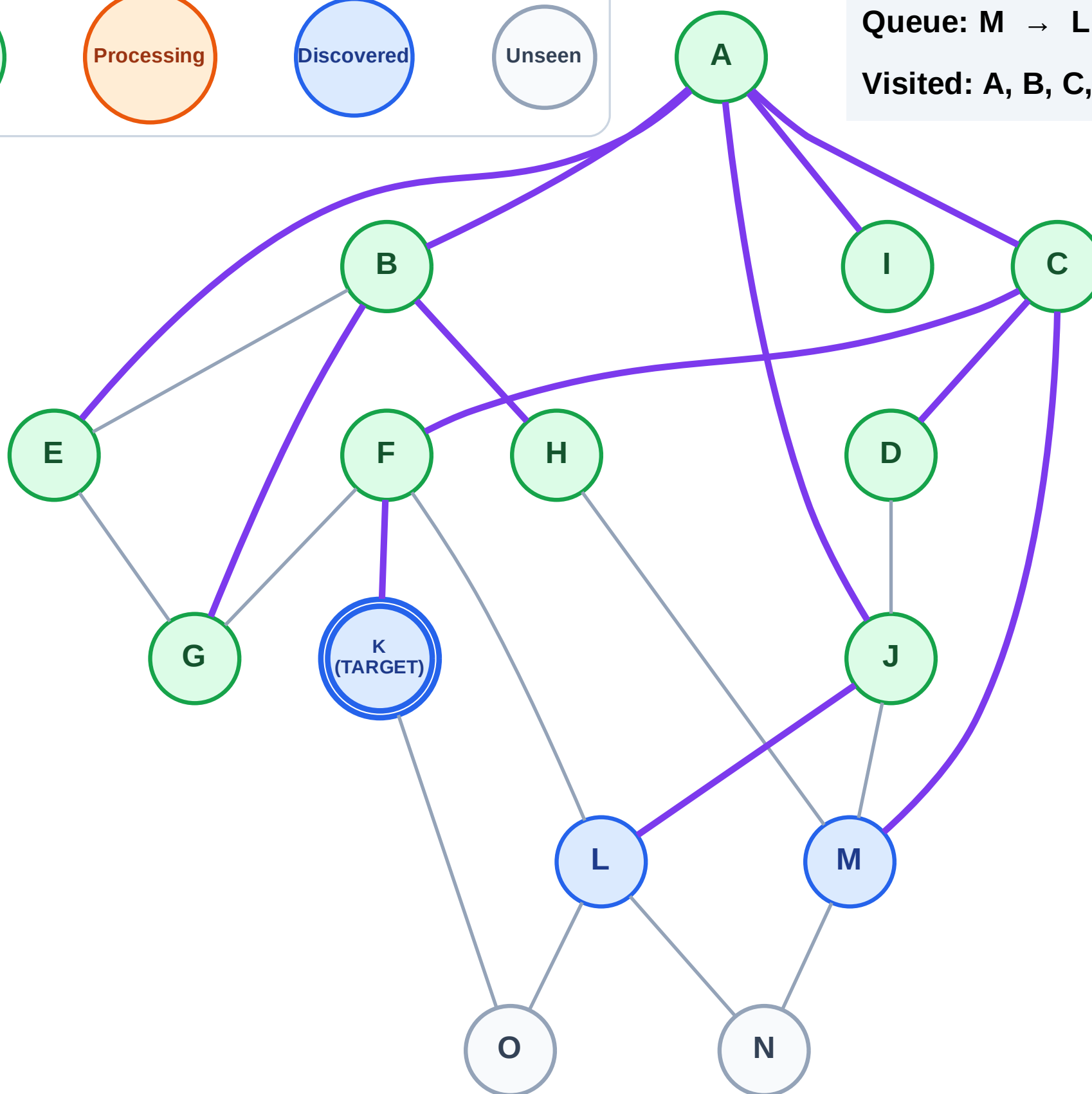
Step 21: Discover K from F

Legend



Queue: M → L → K

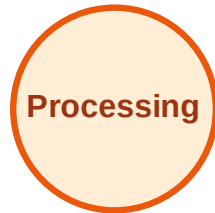
Visited: A, B, C, D, E, F, G, H, I, J



BFS Traversal

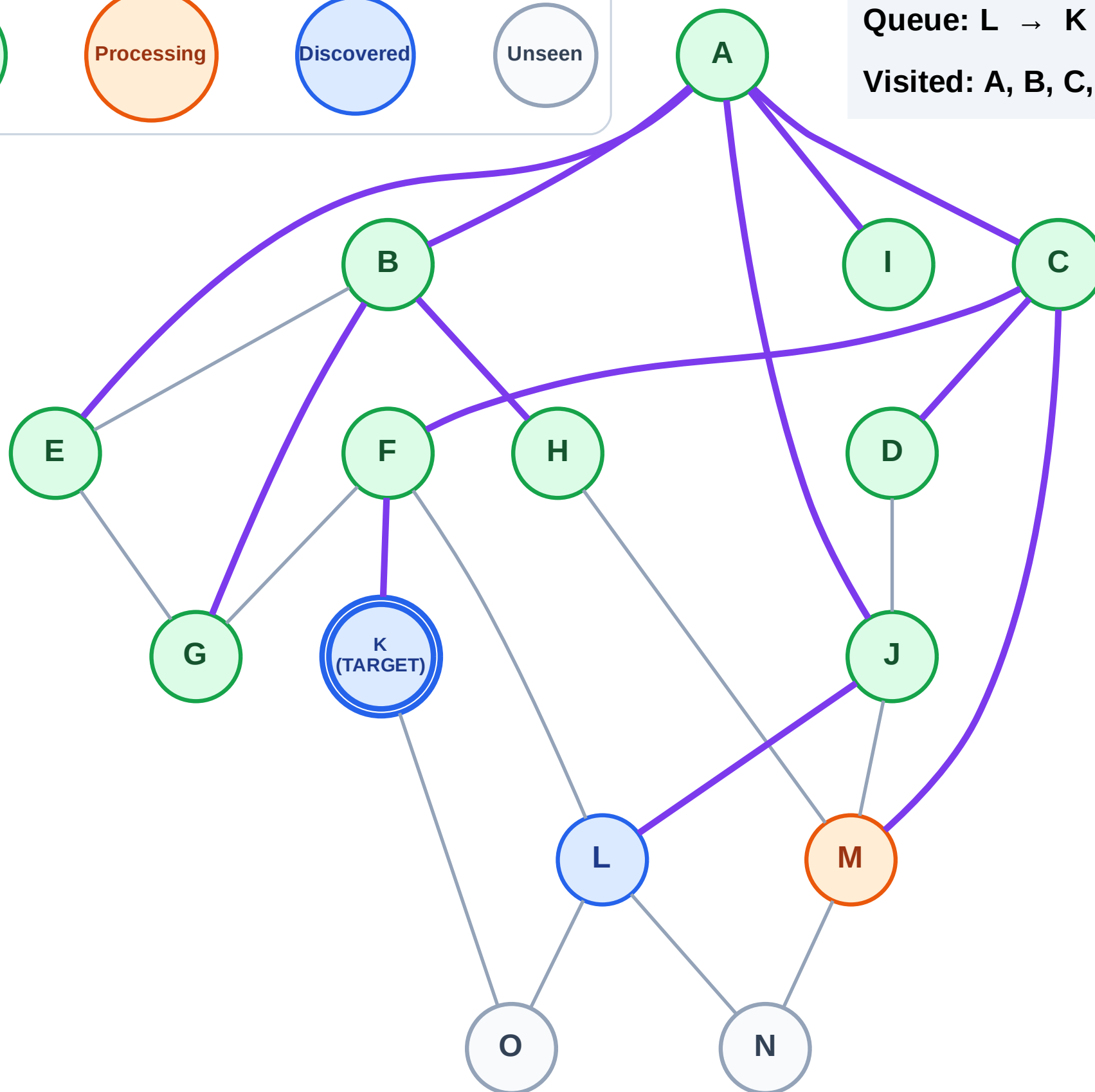
Step 22: Process node M

Legend



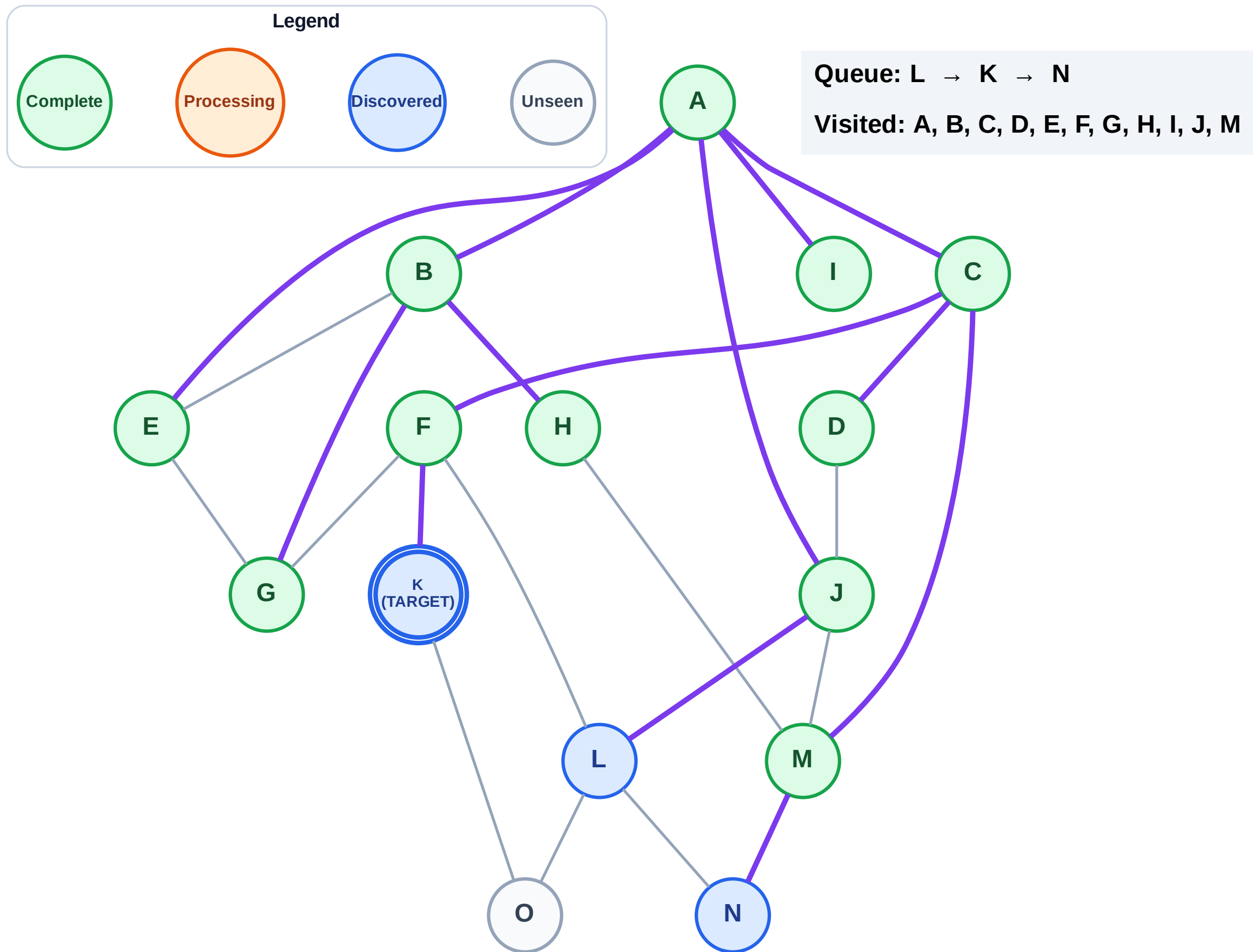
Queue: L → K

Visited: A, B, C, D, E, F, G, H, I, J



Step 23: Discover N from M

Step 23: Discover N from M



BFS Traversal

Step 24: Process node L

Legend

Complete

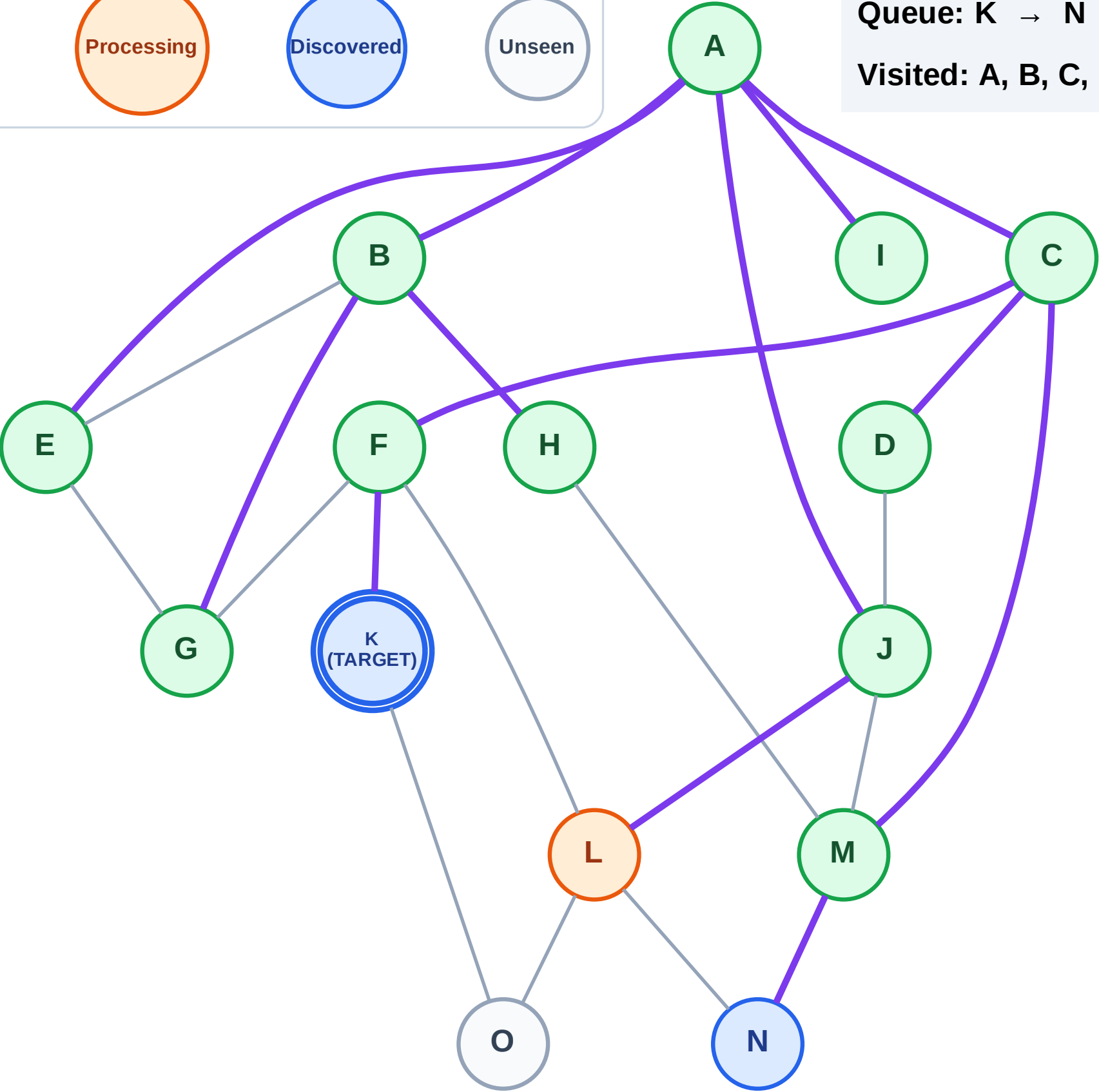
Processing

Discovered

Unseen

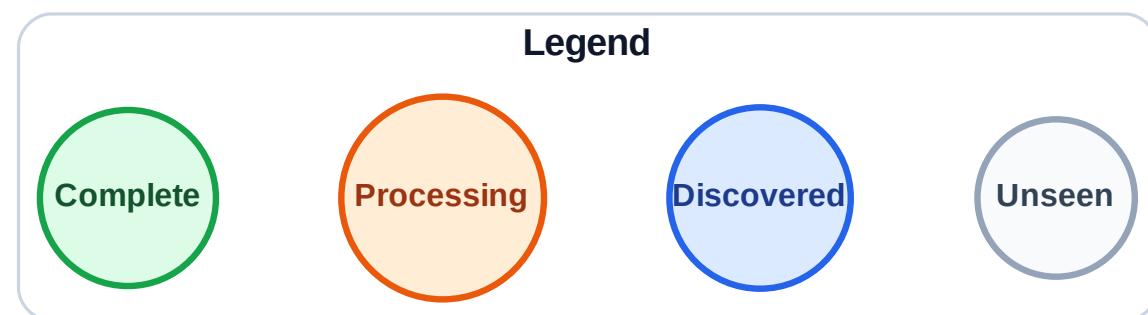
Queue: K → N

Visited: A, B, C, D, E, F, G, H, I, J, M



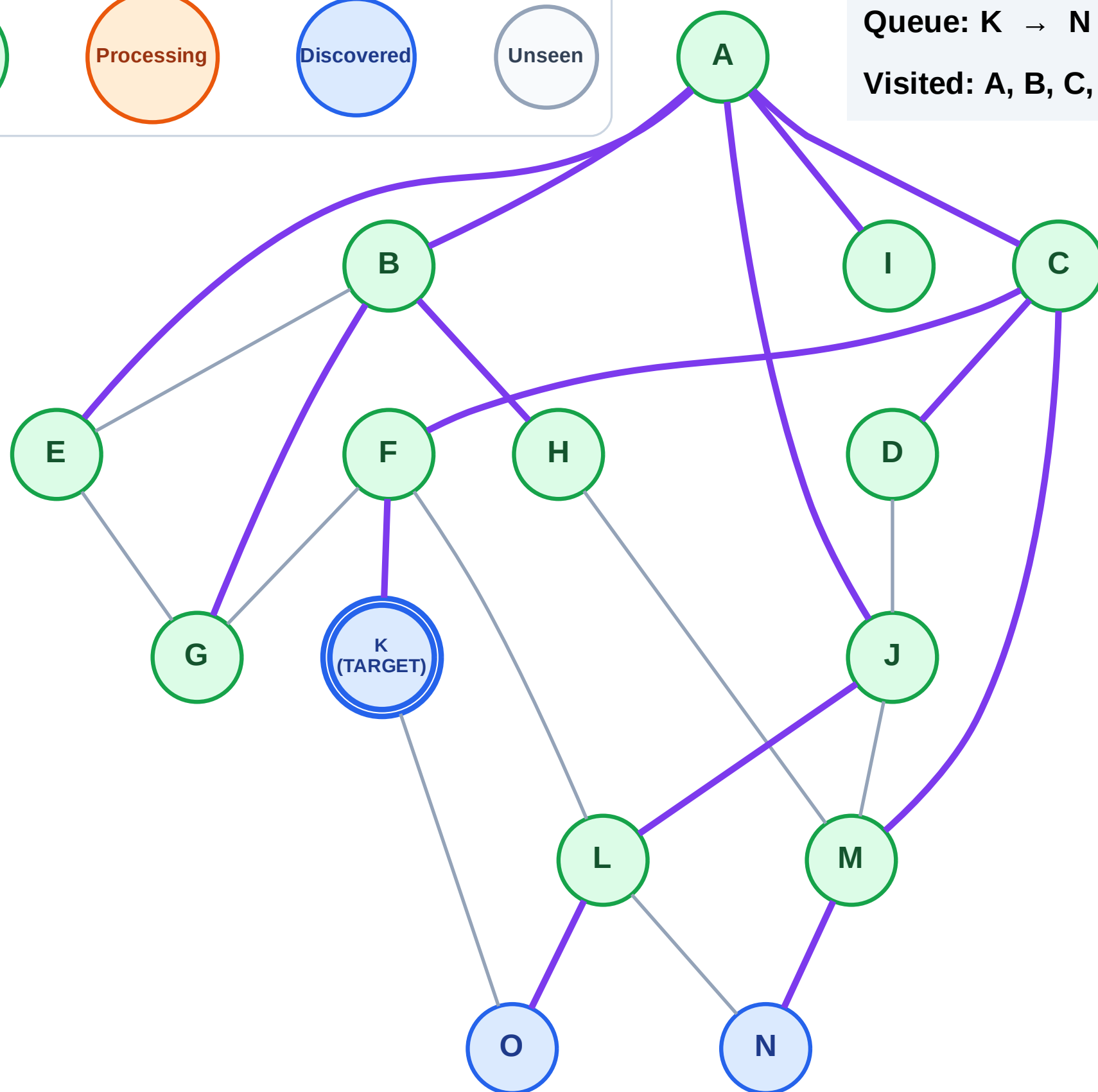
BFS Traversal

Step 25: Discover O from L



Queue: K → N → O

Visited: A, B, C, D, E, F, G, H, I, J, L, M



BFS Traversal

Step 26: Process node K

Legend

Complete

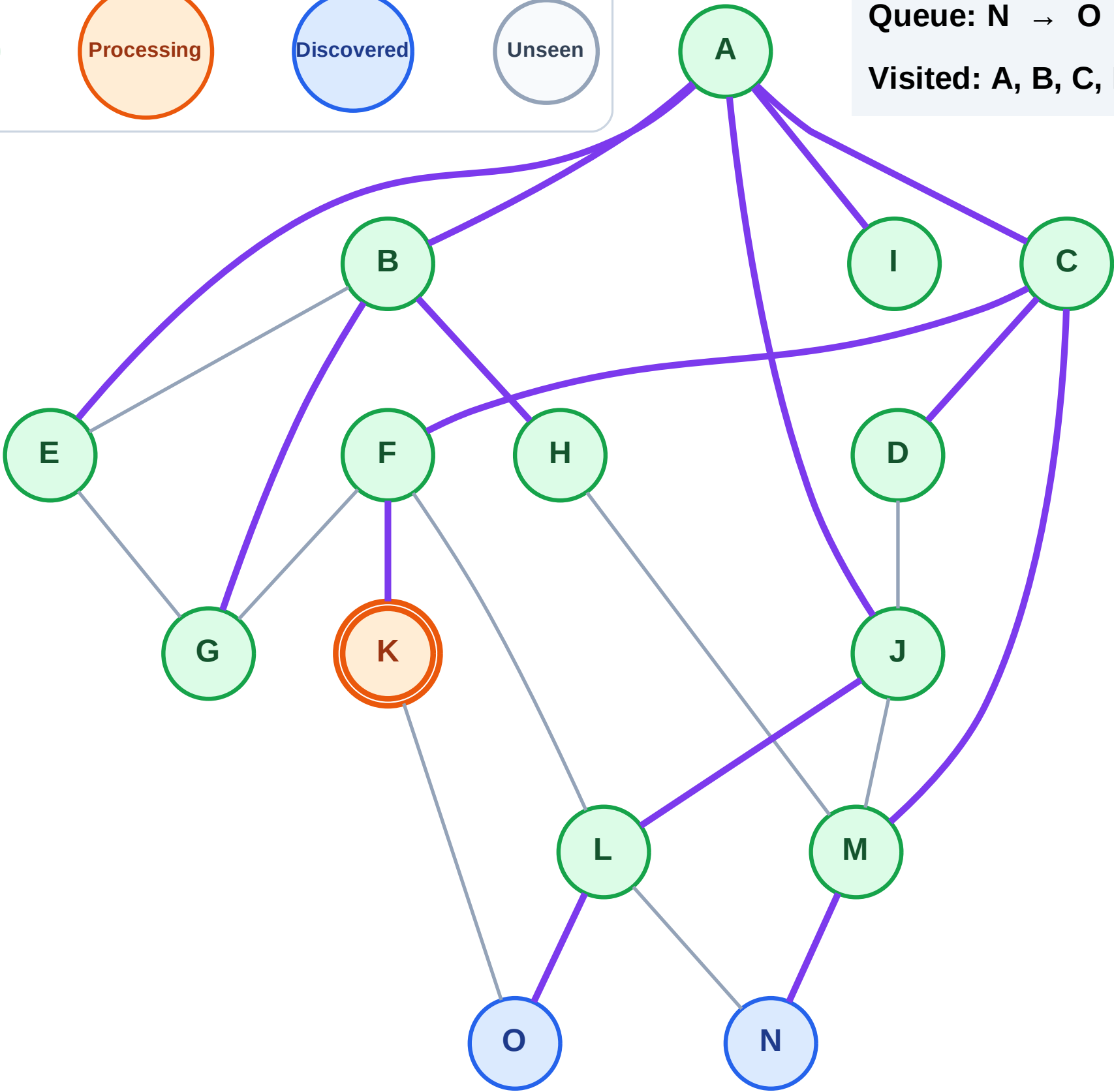
Processing

Discovered

Unseen

Queue: N → O

Visited: A, B, C, D, E, F, G, H, I, J, L, M



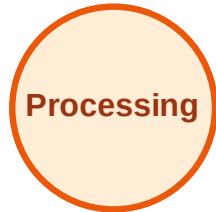
BFS Traversal

Step 27: TARGET NODE K REACHED! Path (3 hops): A → C → F → K

Legend



Complete



Processing



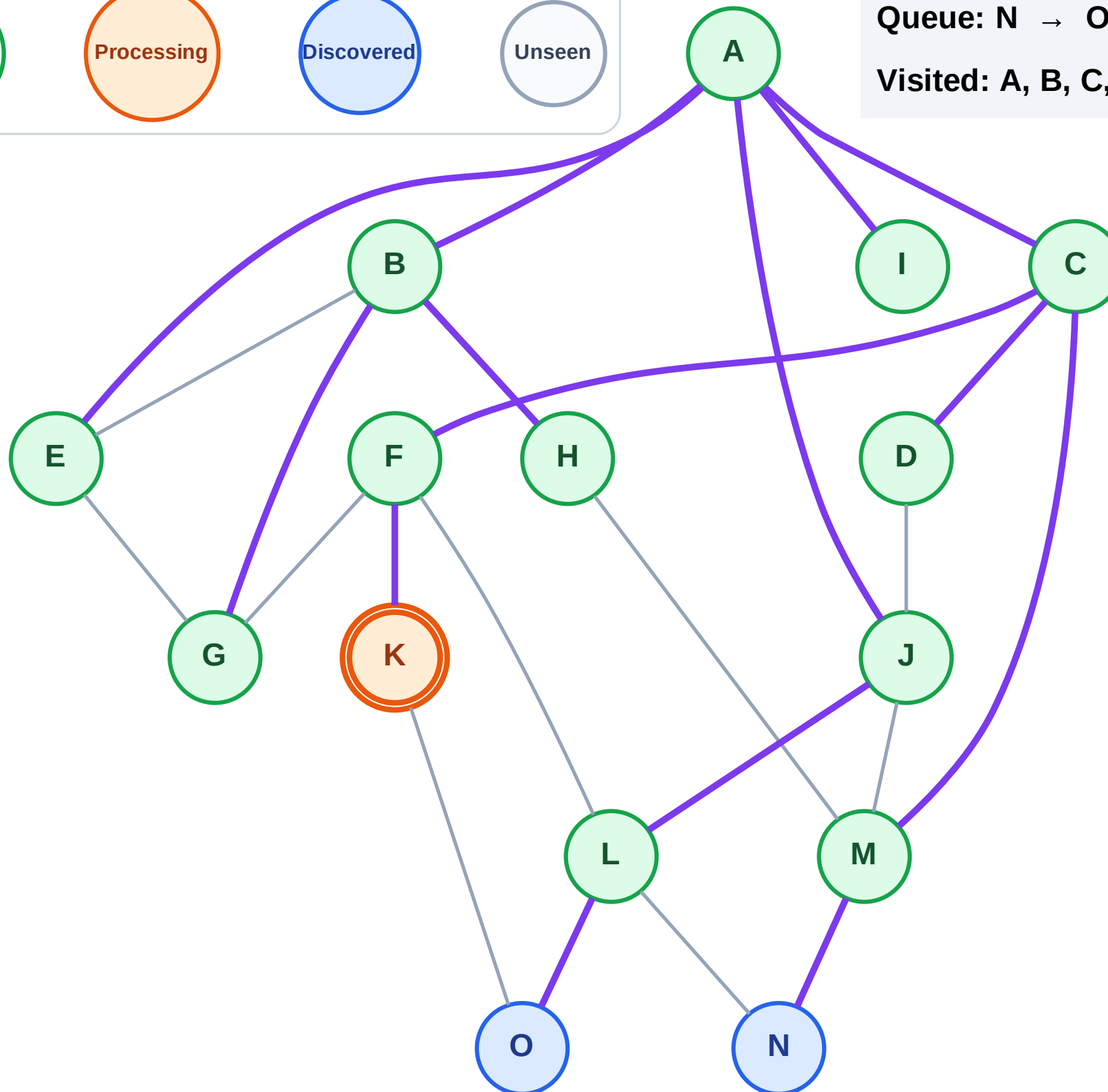
Discovered



Unseen

Queue: N → O

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

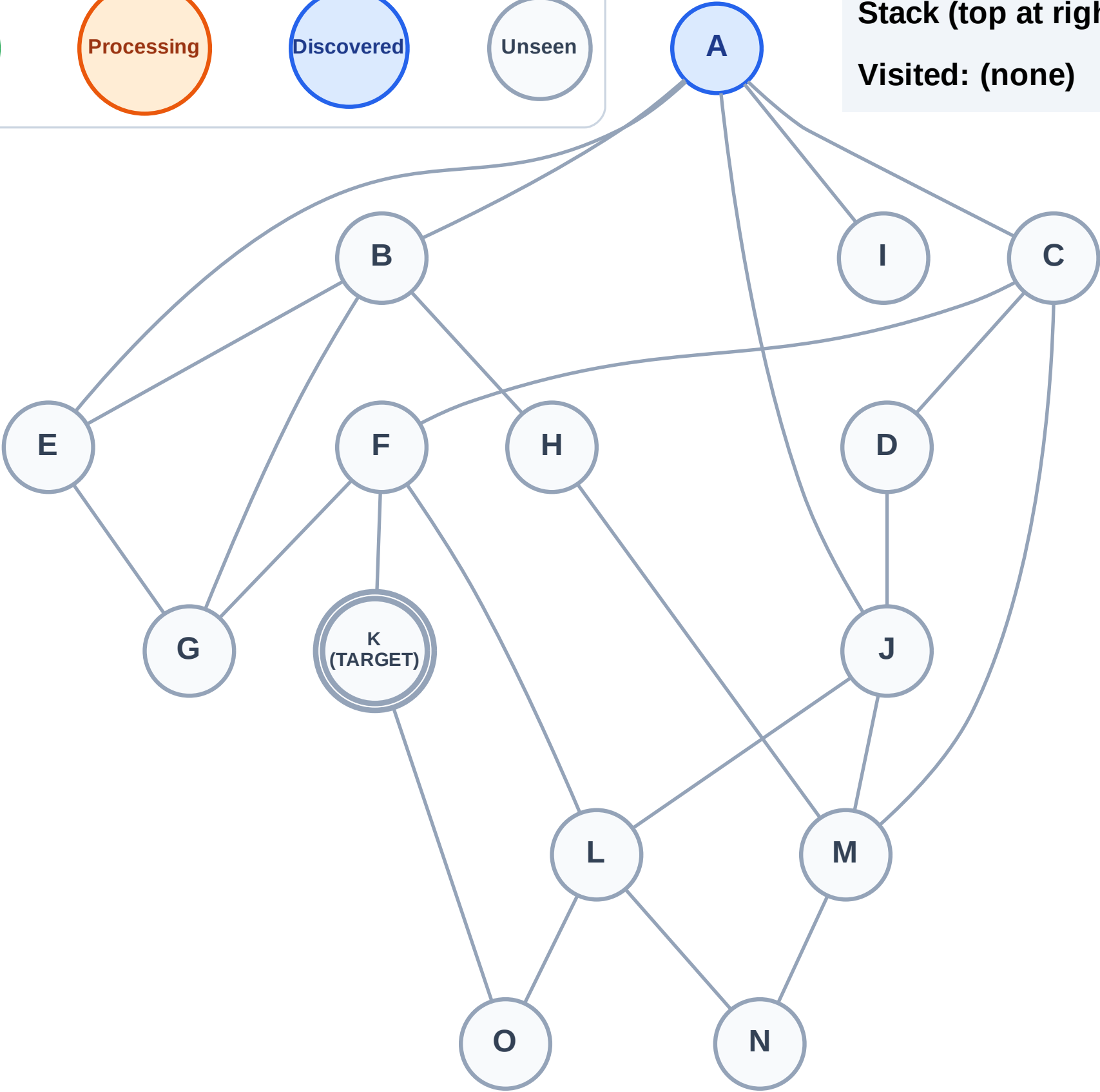
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

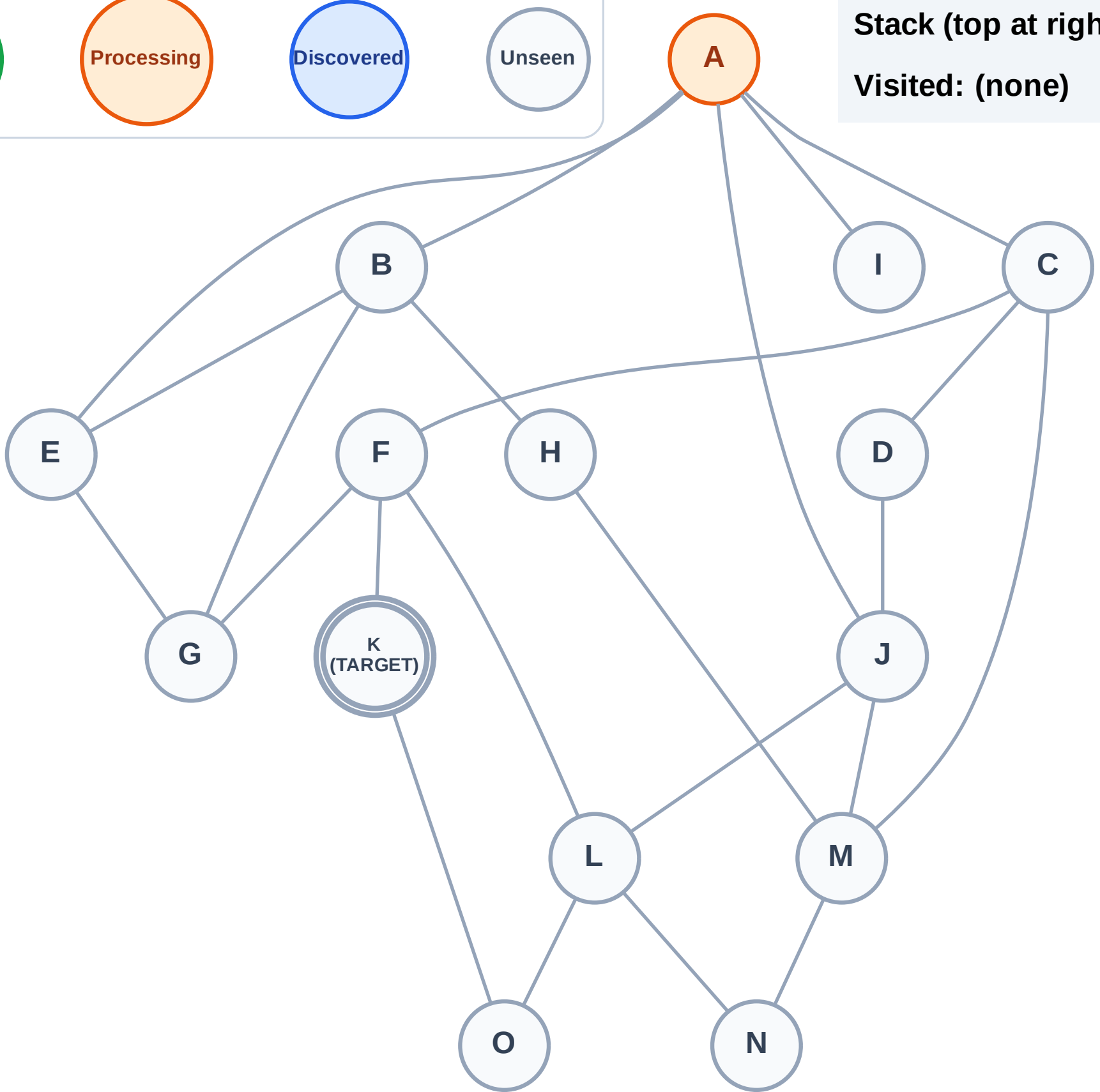
Complete

Processing

Discovered

Unseen

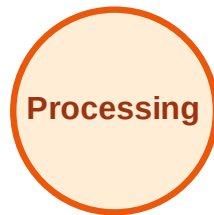
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

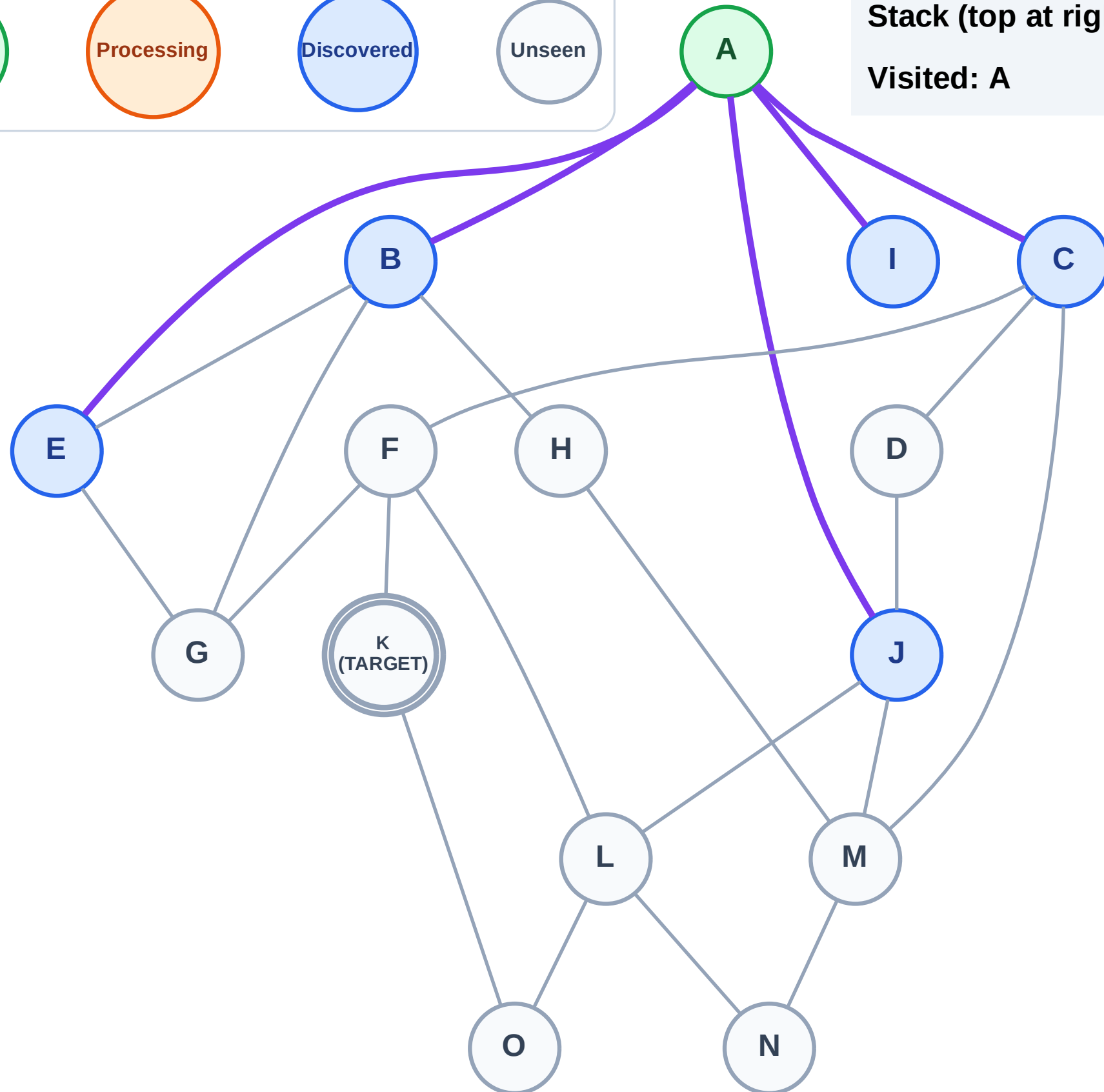
Step 3: Discover J, I, E, C, B from A

Legend



Stack (top at right): J | I | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

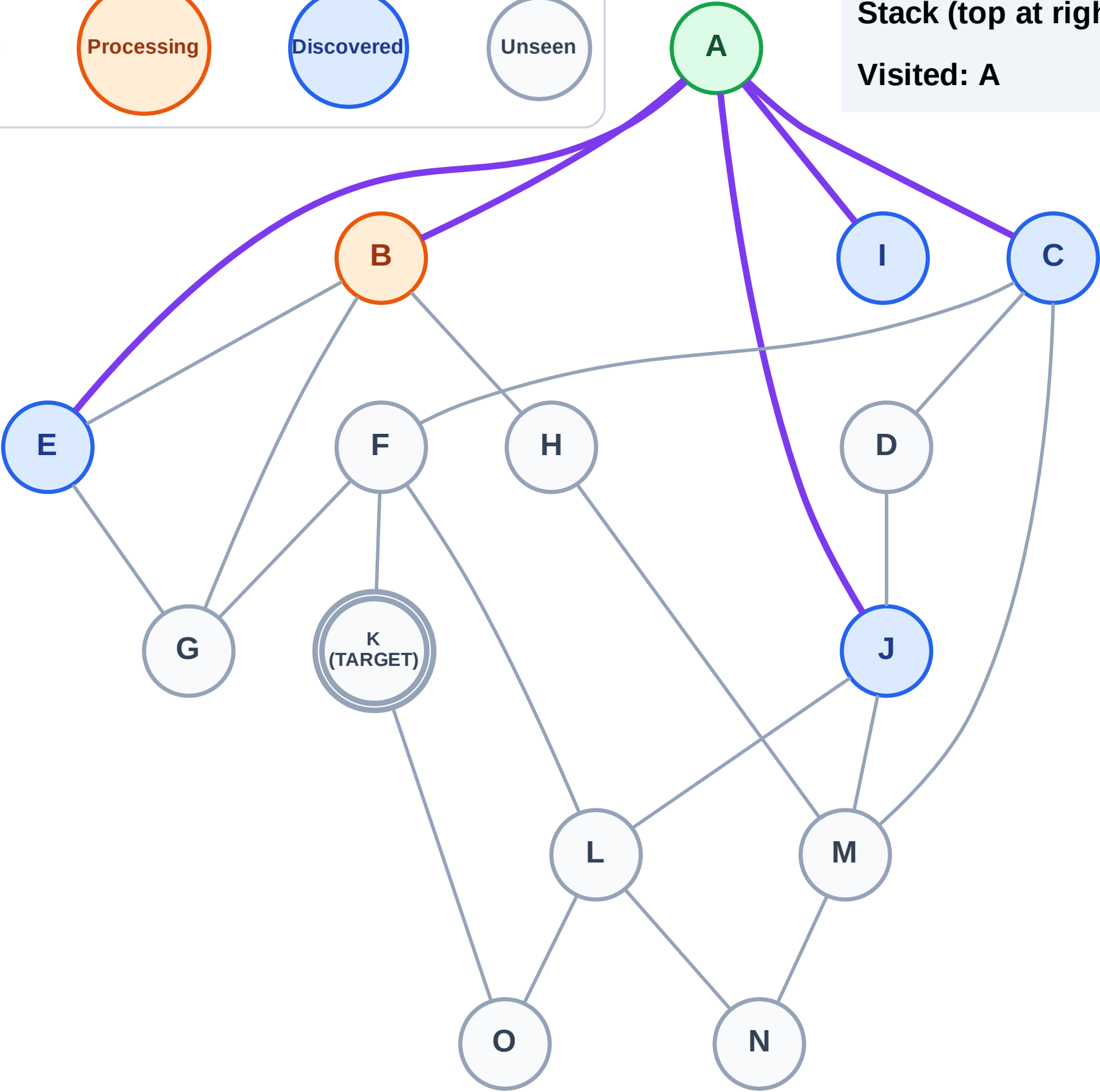
Complete

Processing

Discovered

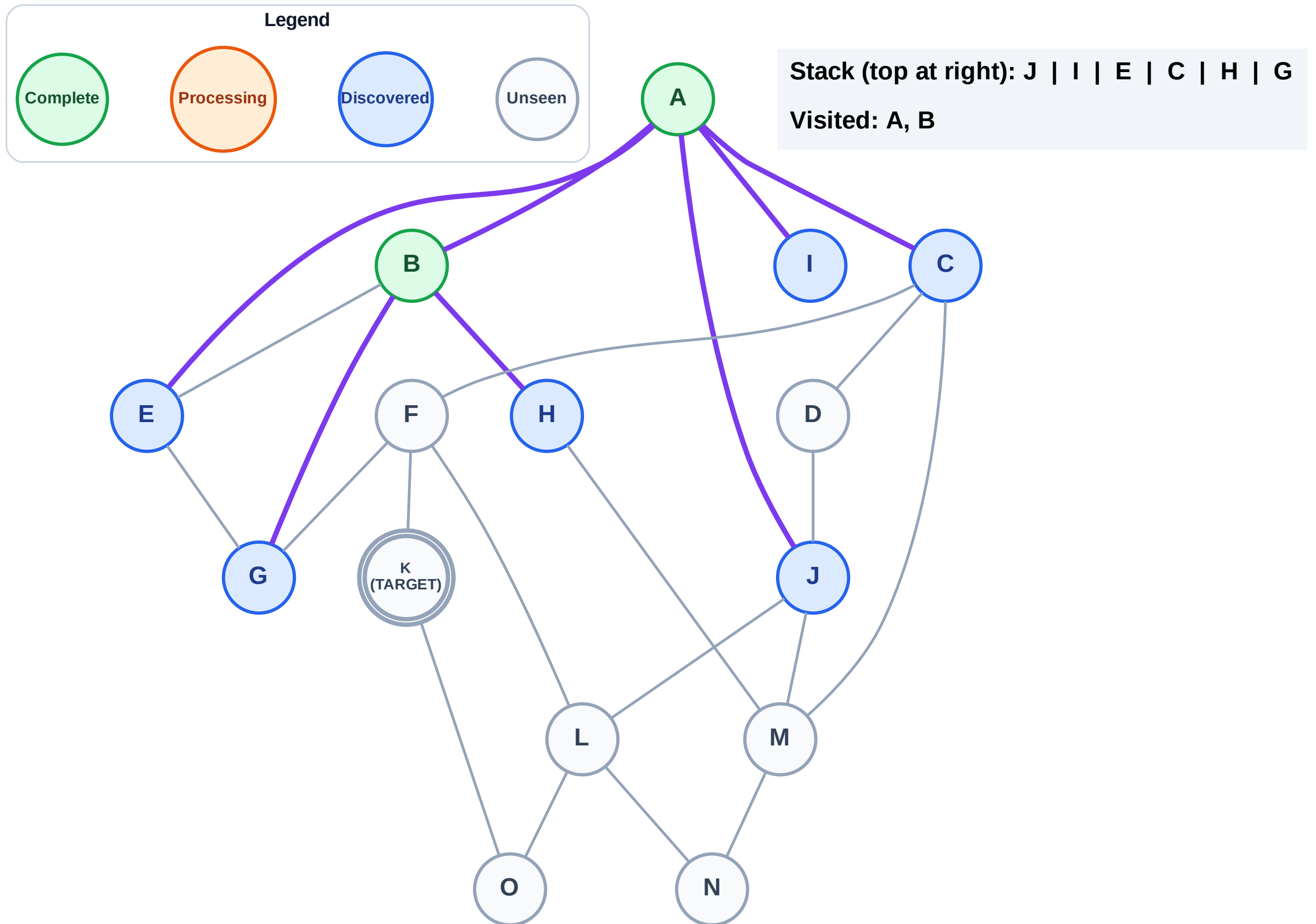
Unseen

Stack (top at right): J | I | E | C
Visited: A



Step 5: Discover H, G from B

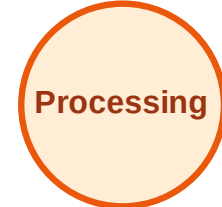
Step 5: Discover H, G from B



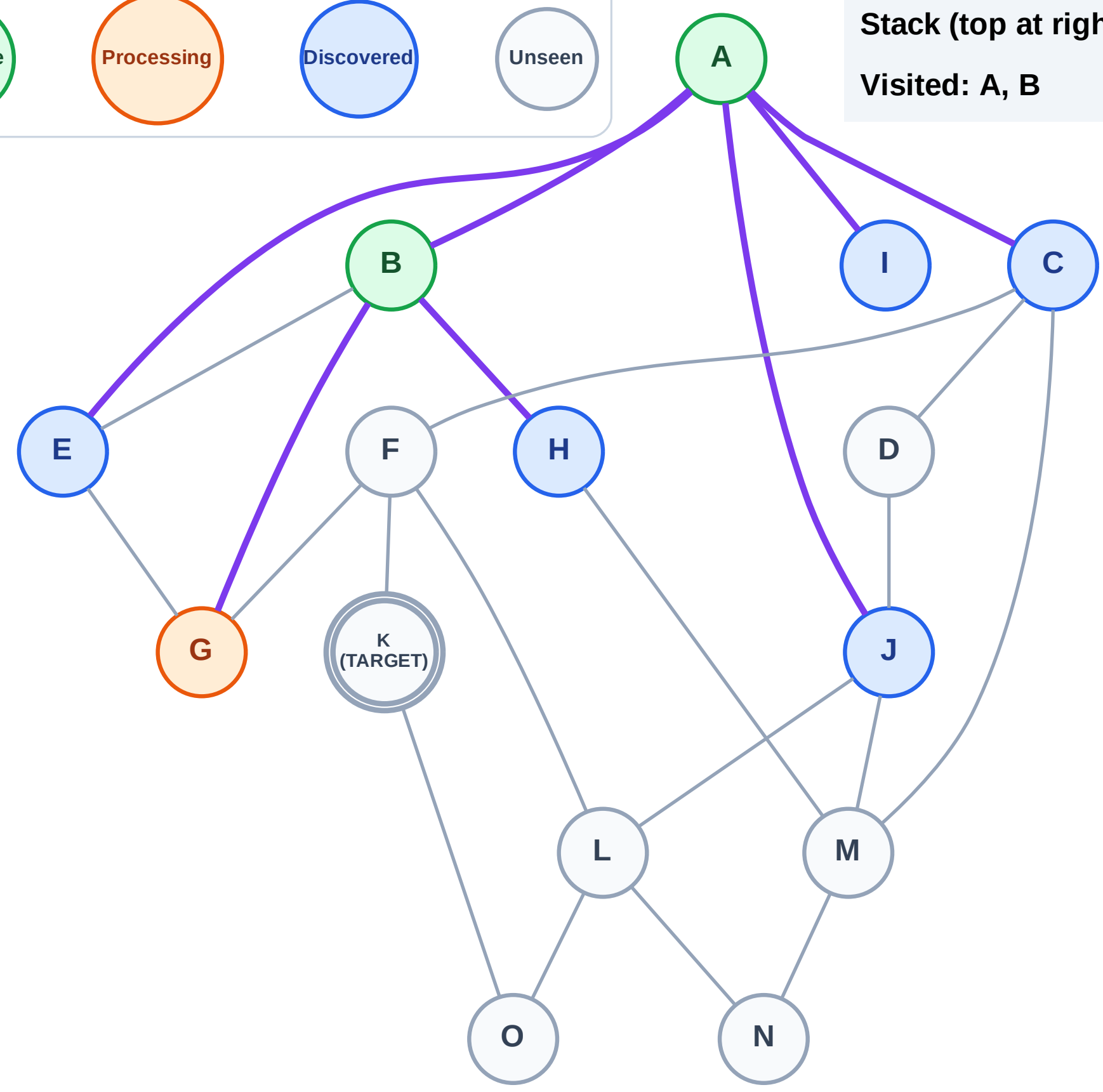
DFS Traversal

Step 6: Process node G

Legend



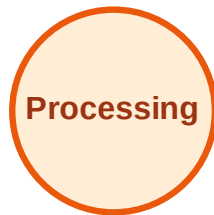
Stack (top at right): J | I | E | C | H
Visited: A, B



DFS Traversal

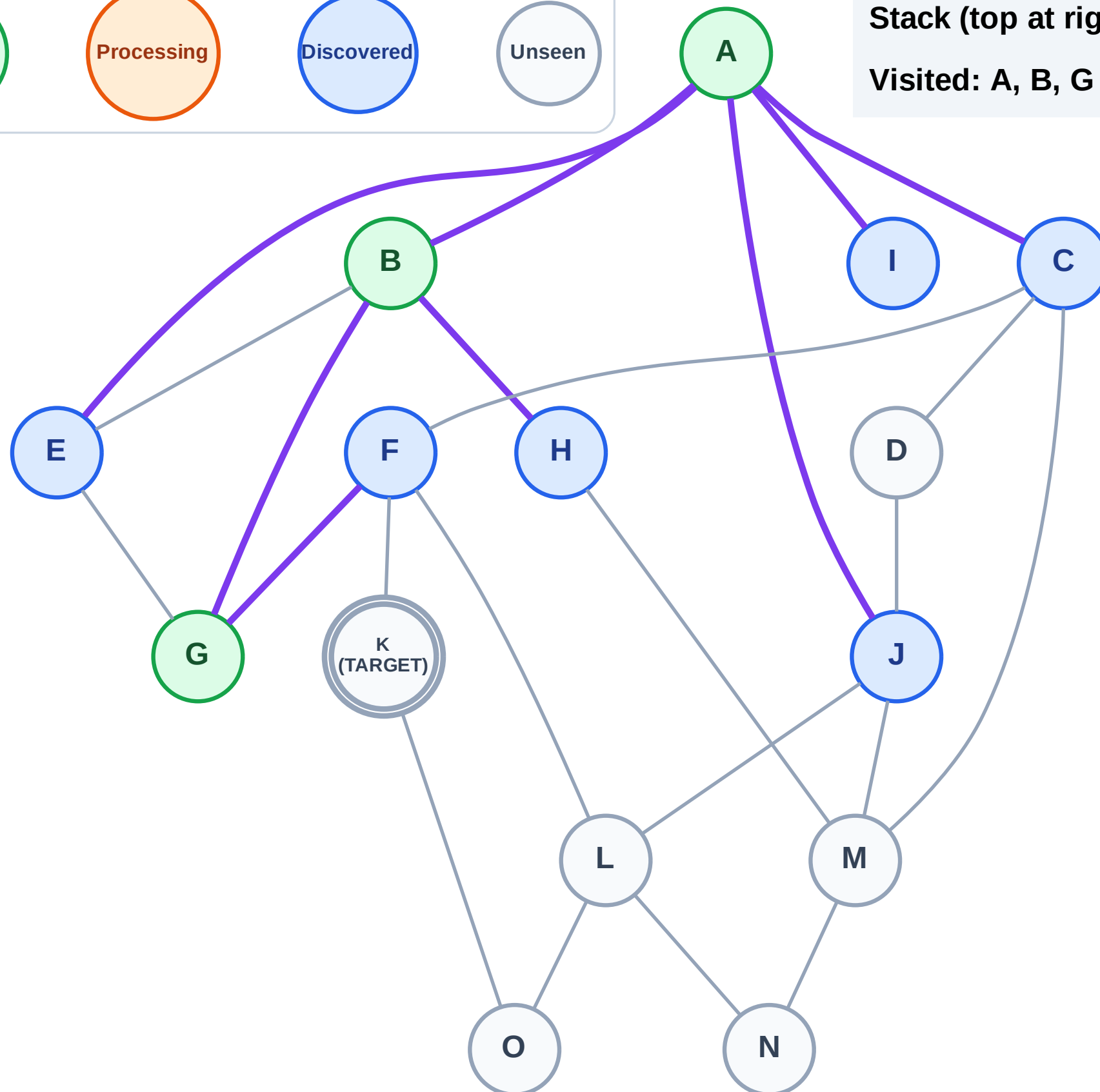
Step 7: Discover F from G

Legend



Stack (top at right): J | I | E | C | H | F

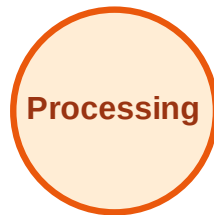
Visited: A, B, G



DFS Traversal

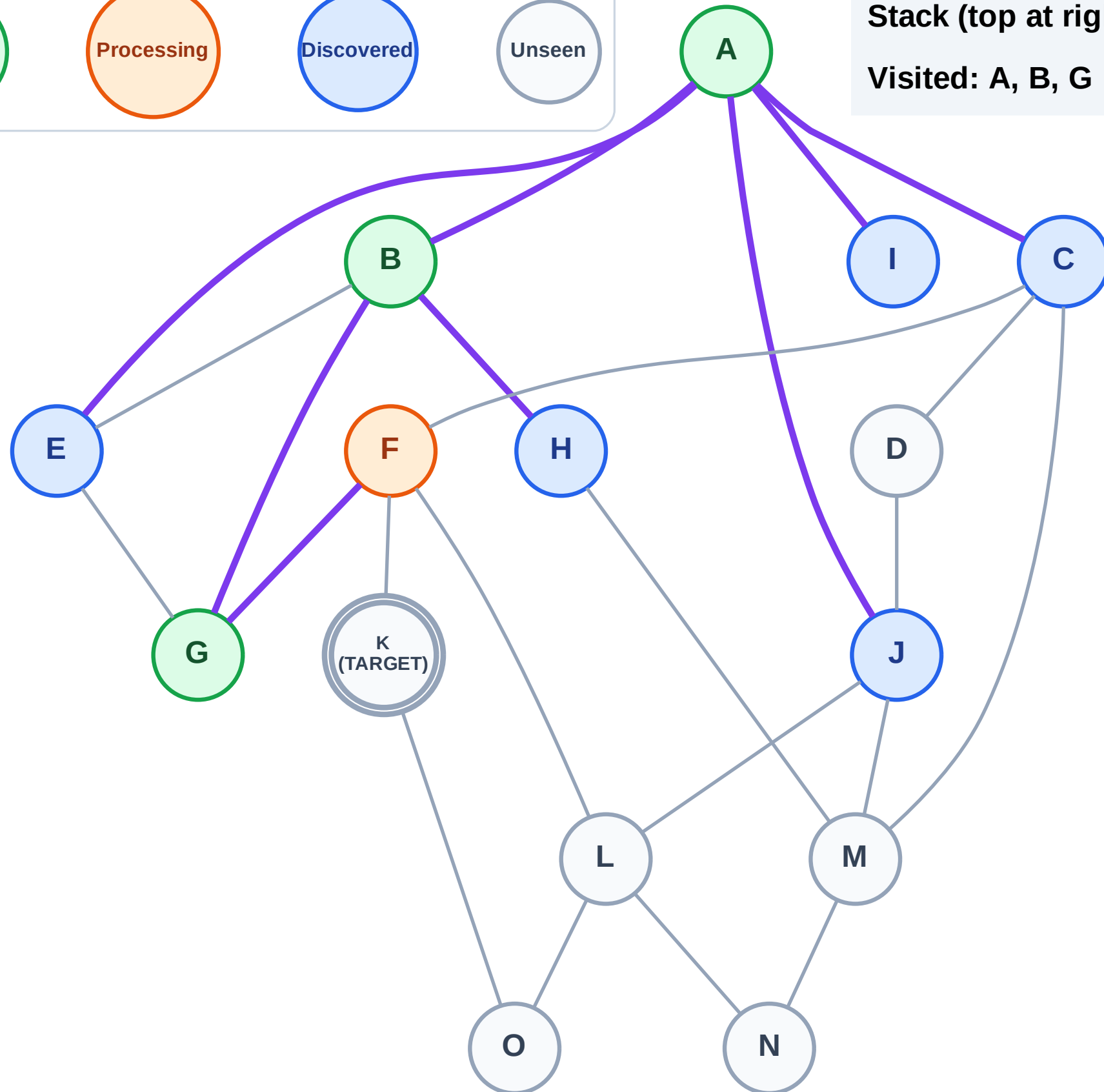
Step 8: Process node F

Legend



Stack (top at right): J | I | E | C | H

Visited: A, B, G



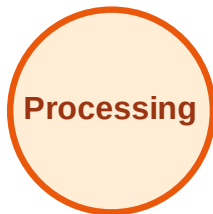
DFS Traversal

Step 9: Discover L, K from F

Legend



Complete



Processing



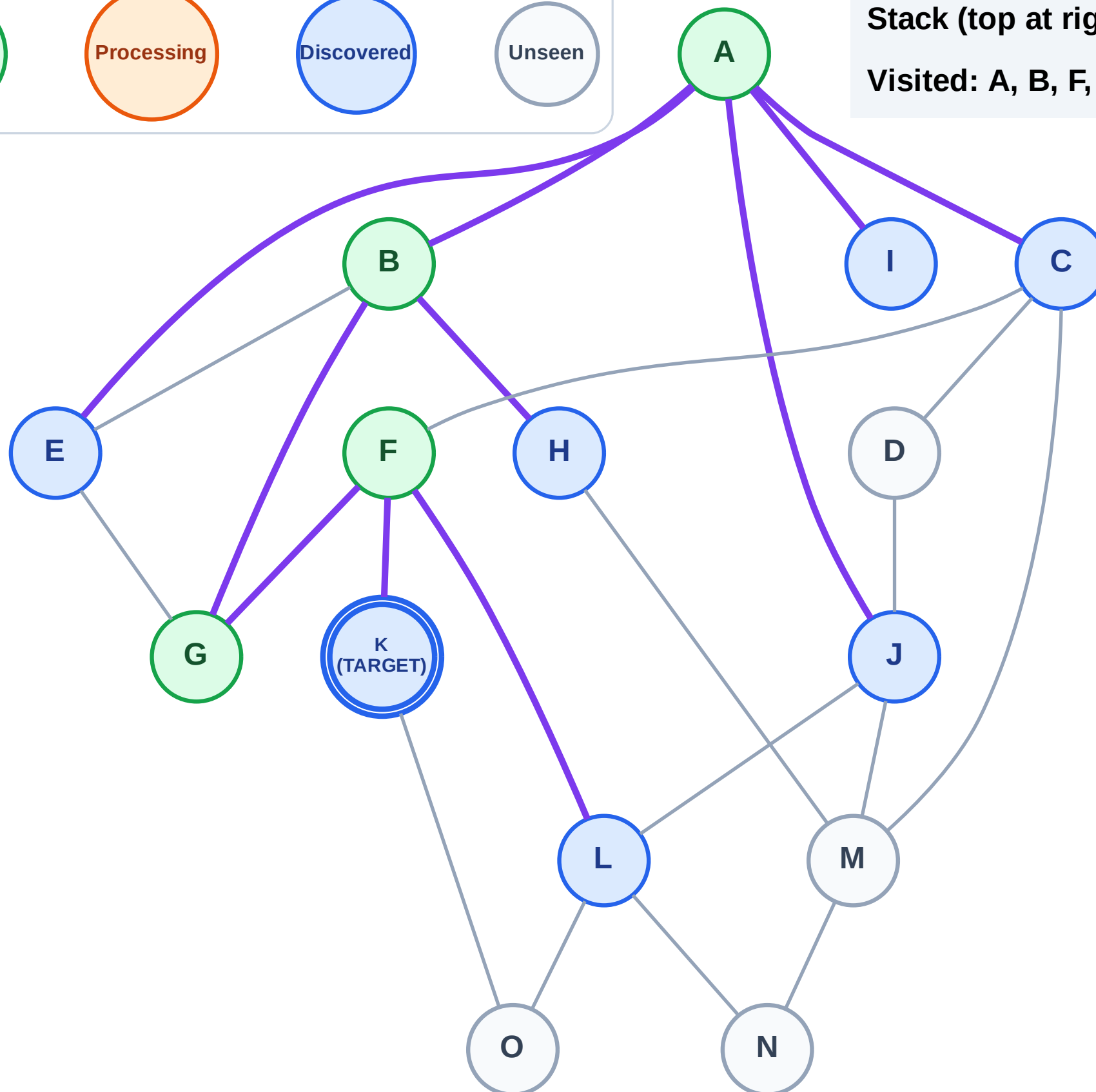
Discovered



Unseen

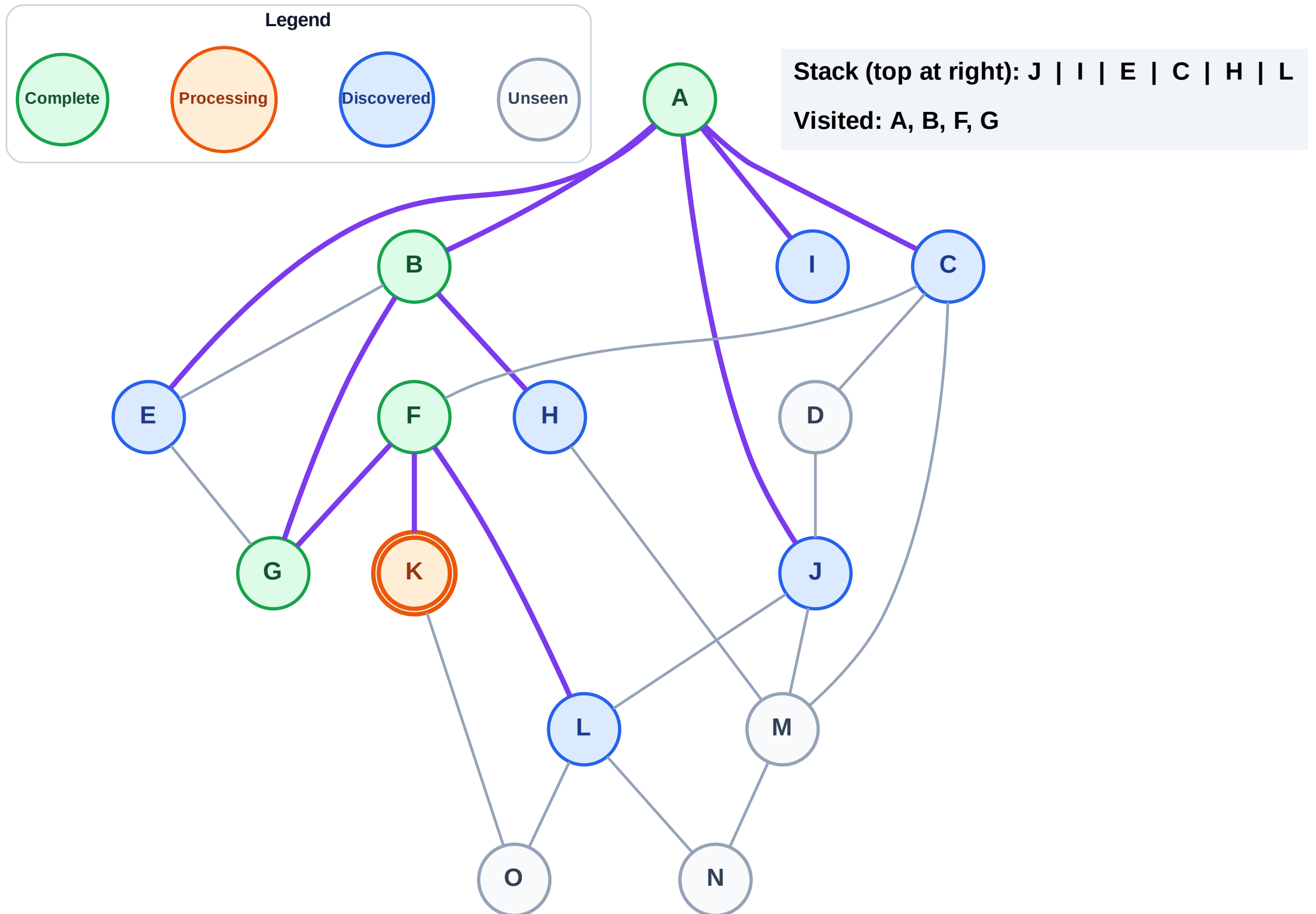
Stack (top at right): J | I | E | C | H | L | K

Visited: A, B, F, G



Step 10: Process node K

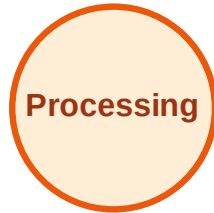
Step 10: Process node K



DFS Traversal

Step 11: TARGET NODE K REACHED! Path (4 hops): A → B → G → F → K

Legend



Stack (top at right): J | I | E | C | H | L

Visited: A, B, F, G, K

